

Riemannian Geometry

Day 1

These notes develop the main definitions and calculations in detail, including clarifications of potentially confusing points.

1 Smooth manifolds and smooth structures

[Definition] A smooth n -manifold is a second-countable Hausdorff space locally homeomorphic to $\mathbb{R}^n$, equipped with a maximal atlas whose transition maps are C^∞ .

The relevant regularity classes are:

- C^0 : topological manifolds;
- C^k : differentiable manifolds of finite regularity;
- C^∞ : smooth manifolds;
- C^ω : real-analytic manifolds.

For a C^k manifold, every transition map has continuous derivatives through order k . Derivatives of order greater than k need not exist. For a C^ω manifold, every transition map is locally represented by a convergent real power series. Therefore

$$C^\omega \subset C^\infty \subset C^k$$

for every finite k . The inclusions are strict. A smooth function need not be analytic because its Taylor series may fail to recover the function. For example, the function equal to e^{-1/x^2} for $x \neq 0$ and equal to zero at $x = 0$ is smooth but not analytic at the origin.

Whitney's smoothing theorem says that a C^1 manifold admits a compatible C^∞ structure. In lower regularity, such a smoothing need not exist in every dimension, and a fixed topological manifold may admit inequivalent smooth structures.

A smooth manifold also admits a compatible real-analytic structure, but there is no canonical analytic atlas. The appropriate uniqueness statement is only up to real-analytic diffeomorphism compatible with the underlying smooth structure, not literal equality of atlases.

Exotic smooth structures are different smooth structures on the same underlying topological manifold which are not related by a diffeomorphism. Milnor's exotic 7-spheres are the classical examples.

Whether the topological 4-sphere admits an exotic smooth structure is the smooth four-dimensional Poincare problem and remains unresolved. This is exceptional: exotic spheres are understood in dimensions ≥ 5 , while dimension four has special phenomena.

2 Germs, tangent vectors, and the tangent bundle

Germs

[Definition] The germ of a smooth function at $p \in M$ records only its behaviour in an arbitrarily small neighbourhood of p . Two functions f and g define the same germ if they agree on some neighbourhood of p .

[Definition] The algebra of germs is denoted $\mathcal{E}_p$. A tangent vector may be defined as a derivation

$$v : \mathcal{E}_p \rightarrow \mathbb{R}$$

satisfying linearity and the Leibniz rule

$$v(fg) = f(p)v(g) + g(p)v(f).$$

This definition is coordinate-free.

Curve definition

Alternatively, let $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ be smooth with $\gamma(0) = p$. Two such curves are equivalent if

$$\left. \frac{d}{dt}(f \circ \gamma_1)(t) \right|_{t=0} = \left. \frac{d}{dt}(f \circ \gamma_2)(t) \right|_{t=0}$$

for every smooth f . The equivalence class $[\gamma]$ defines the derivation

$$v(f) = \left. \frac{d}{dt}(f \circ \gamma)(t) \right|_{t=0}.$$

Thus tangent vectors are first-order velocities.

Coordinates

In a chart $(U, x^1, \dots, x^n)$, the coordinate vectors

$$\left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p$$

form a basis of $T_p M$. Under a coordinate change $y = y(x)$,

$$\frac{\partial}{\partial x^i} = \sum_j \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j}.$$

The tangent bundle

$$TM = \bigsqcup_{p \in M} T_p M$$

is itself a smooth $2n$ -manifold. Local coordinates give bundle charts

$$TM|_U \cong U \times \mathbb{R}^n,$$

and the Jacobian matrices of coordinate changes are the transition functions.

3 Vector bundles and sections

[Definition] A rank- r smooth vector bundle is a triple (E, B, π) where E is the total space, B the base, and $\pi : E \rightarrow B$ a smooth surjection such that each fibre

$$E_b = \pi^{-1}(b)$$

is an r -dimensional vector space and each point has a neighbourhood U with a fibrewise-linear diffeomorphism

$$\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r.$$

On overlaps, trivializations differ by a smooth transition map

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow GL(r, \mathbb{R}).$$

[Definition] A section is a smooth map $s : B \rightarrow E$ satisfying $\pi \circ s = \text{id}_B$. The space of smooth sections is denoted $\Gamma(E)$. For the tangent bundle, $\Gamma(TM)$ is the space of smooth vector fields.

4 Connections

[Definition] A connection on a vector bundle $E \rightarrow M$ is a map

$$\nabla : \Gamma(TM) \times \Gamma(E) \rightarrow \Gamma(E), \quad (X, s) \mapsto \nabla_X s,$$

satisfying

$$\nabla_{fX} s = f \nabla_X s,$$

$$\nabla_{X+Y} s = \nabla_X s + \nabla_Y s,$$

$$\nabla_X (fs) = X(f)s + f \nabla_X s,$$

and linearity in the section variable. A connection differentiates a section along a direction even though values of the section lie in different fibres.

5 Riemannian metrics

[Definition] A Riemannian metric g assigns to each $p \in M$ a positive-definite inner product g_p on $T_p M$, smoothly in p .

Equivalently, it acts on smooth vector fields as a map

$$g : \Gamma(TM) \times \Gamma(TM) \longrightarrow C^\infty(M),$$

where the resulting smooth function is defined pointwise by

$$g(X, Y)(p) = g_p(X_p, Y_p).$$

The codomain here is $C^\infty(M)$ rather than $\mathbb{R}$. The pointwise metric g_p has values in $\mathbb{R}$. The map on vector fields is bilinear over smooth functions:

$$g(fX + hY, Z) = f g(X, Z) + h g(Y, Z)$$

for all $f, h \in C^\infty(M)$. This $C^\infty(M)$ -bilinearity implies that $g(X, Y)(p)$ depends only on the vectors X_p and Y_p , not on how the vector fields are extended away from p . Therefore g is a smooth covariant 2-tensor,

$$g \in \Gamma(T^*M \otimes T^*M).$$

This justifies tensor notation. In local coordinates,

$$g = g_{ij} dx^i \otimes dx^j, \quad g_{ij}(p) = g_p(\partial_i, \partial_j),$$

where (g_{ij}) is symmetric positive definite. The norm is $|v|_g = \sqrt{g(v, v)}$.

Every smooth manifold admits a Riemannian metric. Choose local Euclidean metrics g_α and a partition of unity $\{\varphi_\alpha\}$ subordinate to the coordinate cover; then

$$g = \sum_{\alpha} \varphi_{\alpha} g_{\alpha}$$

is globally defined and positive definite.

Pullback metrics and isometric immersions

[Definition] For a smooth map $F : M \rightarrow N$ and a metric h on N , define

$$(F^*h)_p(v, w) = h_{F(p)}(dF_p v, dF_p w).$$

This is positive definite exactly when dF_p is injective, so an immersion pulls back a Riemannian metric. The map is an isometric immersion when

$$F^*h = g.$$

Example: the standard round metric on

$$S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$$

is the pullback of the Euclidean metric under the inclusion $S^n \hookrightarrow \mathbb{R}^{n+1}$.

Product and warped-product metrics

[Example] On $M_1 \times M_2$, the product metric is

$$g = \pi_1^* g_1 \oplus \pi_2^* g_2.$$

For $v = (v_1, v_2)$ and $w = (w_1, w_2)$,

$$g(v, w) = g_1(v_1, w_1) + g_2(v_2, w_2).$$

If $f : M_1 \rightarrow (0, \infty)$ is smooth, the warped product is

$$g = g_1 \oplus f^2 g_2.$$

6 Riemannian coverings and quotient metrics

[Definition] If $\pi : \widetilde{M} \rightarrow M$ is a covering map and (M, g) is Riemannian, the pullback metric

$$\widetilde{g} = \pi^* g$$

makes π a local isometry. The pair $(\widetilde{M}, \widetilde{g})$ is a Riemannian covering space.

For the antipodal quotient $\pi : S^2 \rightarrow \mathbb{RP}^2$, the round metric descends.

The round metric is invariant under the antipodal map $A(x) = -x$, because $dA = -I$ is orthogonal. Hence $A^* g_{S^2} = g_{S^2}$, so there is a unique metric on $\mathbb{RP}^2$ for which π is a local isometry.

More generally, a properly discontinuous group action by isometries produces a quotient metric.

Every deck transformation of a Riemannian covering is an isometry: if γ is a deck transformation, $\pi \circ \gamma = \pi$, hence $\gamma^* \widetilde{g} = \gamma^* \pi^* g = (\pi \circ \gamma)^* g = \pi^* g = \widetilde{g}$.

7 Riemannian submersions

[Definition] Let $\pi : (M, g) \rightarrow (N, h)$ be a submersion. The vertical subspace is

$$V_p = \ker d\pi_p,$$

and the horizontal subspace is its orthogonal complement

$$H_p = V_p^\perp, \quad T_p M = V_p \oplus H_p.$$

The map is a Riemannian submersion if

$$d\pi_p|_{H_p} : H_p \rightarrow T_{\pi(p)}N$$

is an isometry. Equivalently,

$$g|_{H_p} = \pi^*h|_{H_p}.$$

The full pullback π^*h is only semidefinite on TM because it vanishes on vertical vectors; this explains why a submersion does not simply pull a metric back to a Riemannian metric on all of M .

8 Levi-Civita connection

Fundamental theorem

[Theorem] On every Riemannian manifold (M, g) there exists a unique connection ∇ satisfying:

1. metric compatibility,

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z);$$

1. zero torsion,

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

This is the Levi-Civita connection.

Koszul formula

[Corollary] Combining cyclic permutations of metric compatibility and using zero torsion gives the Koszul formula

$$\begin{aligned} 2g(\nabla_X Y, Z) &= Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) \\ &\quad + g([X, Y], Z) - g([Y, Z], X) + g([Z, X], Y). \end{aligned}$$

The right side depends only on X, Y, Z and g , so it proves uniqueness; nondegeneracy of g defines $\nabla_X Y$, proving existence.

Christoffel symbols

[Corollary] In local coordinates,

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k,$$

where

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}).$$

The Christoffel symbols are not the components of a tensor: under coordinate changes they acquire an inhomogeneous second-derivative term. The difference of two connections is a tensor because those inhomogeneous terms cancel.

For $X = X^i \partial_i$ and $Y = Y^j \partial_j$,

$$\nabla_X Y = X^i \left(\partial_i Y^k + \Gamma_{ij}^k Y^j \right) \partial_k.$$

Although this formula uses a local extension of a tangent vector, $(\nabla_X Y)_p$ depends only on X_p and the first-order germ of Y at p .

Covariant differentiation along curves and parallel transport

[Definition] Let $\gamma : [a, b] \rightarrow M$ be a smooth curve and let $V(t) \in T_{\gamma(t)} M$ be a vector field along it. Choose any local extension $\tilde{V}$ of V near the curve and define

$$\frac{DV}{dt} = \nabla_{\dot{\gamma}} \tilde{V}.$$

Any extension is valid. If two extensions agree along γ , their difference W vanishes there. In local coordinates, linearity and the connection formula give

$$\nabla_{\dot{\gamma}} W = \dot{\gamma}^i \left(\partial_i W^k + \Gamma_{ij}^k W^j \right) \partial_k.$$

Along the curve, $W^j = 0$ and

$$\dot{\gamma}^i \partial_i W^k = \frac{d}{dt} W^k(\gamma(t)) = 0.$$

Thus $\nabla_{\dot{\gamma}} W = 0$ along γ , so the definition is independent of the chosen extension.

A vector field is parallel when

$$\frac{DV}{dt} = 0.$$

Writing $V = V^k \partial_k$, the parallel-transport equation is

$$\frac{dV^k}{dt} + \Gamma_{ij}^k(\gamma(t)) \dot{\gamma}^i V^j = 0.$$

Existence and uniqueness for this linear ODE gives a linear isomorphism

$$P_\gamma : T_{\gamma(a)} M \rightarrow T_{\gamma(b)} M.$$

Metric compatibility implies that parallel transport preserves inner products:

$$\frac{d}{dt}g(V, W) = g(\nabla_{\dot{\gamma}}V, W) + g(V, \nabla_{\dot{\gamma}}W) = 0.$$

Therefore P_γ is an orthogonal linear map between endpoint tangent spaces.

For a loop based at p , parallel transport gives an element of $O(T_pM, g_p)$, but it need not be the identity.

The identification around a loop is the identity only in special circumstances. Curvature produces infinitesimal holonomy, and global topology can produce additional holonomy. The collection of all loop transports is the holonomy group $\text{Hol}_p(M, g)$.

Day 2

To extend covariant differentiation from vector fields to tensor fields, we require it to satisfy the product rule for tensor products. This ensures that differentiating a tensor is compatible with its decomposition into tensor factors.

9 Extension of covariant differentiation to tensors

[Notation] We use the convention that a tensor of type (s, r) has s contravariant slots and r covariant slots.

First consider a tensor of type $(s, 0)$, which is a section of $\otimes^s TM$. On an ordinary tensor product of vector fields, define

$$\nabla_X(Z_1 \otimes \cdots \otimes Z_s) = \sum_{i=1}^s Z_1 \otimes \cdots \otimes \nabla_X Z_i \otimes \cdots \otimes Z_s.$$

Extend this definition to every section of $\otimes^s TM$ by linearity and the product rule.

Now let T have general type (s, r) . At each point p , use the canonical identification with a linear map between tensor powers:

$$T_p : \otimes^r T_p M \longrightarrow \otimes^s T_p M.$$

Globally, T is a fibrewise-linear bundle map

$$T : \otimes^r TM \longrightarrow \otimes^s TM.$$

If A is a section of $\otimes^r TM$, define

$$(\nabla_X T)(A) = \nabla_X(T(A)) - T(\nabla_X A).$$

For a decomposable input $A = Z_1 \otimes \cdots \otimes Z_r$, the product rule gives

$$\begin{aligned}
(\nabla_X T)(Z_1 \otimes \cdots \otimes Z_r) &= \nabla_X (T(Z_1 \otimes \cdots \otimes Z_r)) \\
&\quad - \sum_{i=1}^r T(Z_1 \otimes \cdots \otimes \nabla_X Z_i \otimes \cdots \otimes Z_r).
\end{aligned}$$

The first term differentiates the tensor produced by T . The sum corrects for differentiating each tensor factor in the input. When $s = 0$, the output is a function and the first term is ordinary directional differentiation by X .

[Remark] Alternative construction. One may instead define covariant differentiation first on vectors and covectors. Its action on vector fields is already given by the connection, while for a covector field α it is defined by

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y).$$

We then force ∇_X to satisfy the product rule. For a decomposable tensor of type (s, r) ,

$$T = Z_1 \otimes \cdots \otimes Z_s \otimes \alpha^1 \otimes \cdots \otimes \alpha^r,$$

this means differentiating each vector factor and each covector factor once, leaving every other factor unchanged, and summing all the terms. Extending by linearity defines $\nabla_X T$ for every tensor of type (s, r) . This construction gives the same extension as the linear-map definition above.

Instead of fixing X , we may retain the differentiating direction as one additional covariant slot. Define the full covariant derivative ∇T by

$$(\nabla T)(Z_1 \otimes \cdots \otimes Z_r \otimes X) = (\nabla_X T)(Z_1 \otimes \cdots \otimes Z_r)$$

and therefore

$$\nabla : \Gamma(\otimes^s TM \otimes \otimes^r T^*M) \longrightarrow \Gamma(\otimes^s TM \otimes \otimes^{r+1} T^*M).$$

Here the final covariant slot records the direction of differentiation.

[Example] The metric. Since the Riemannian metric is a tensor of type $(0, 2)$,

$$(\nabla_X g)(Y, Z) = X(g(Y, Z)) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z).$$

Metric compatibility of the Levi-Civita connection is precisely

$$\nabla g = 0.$$

10 Operators related to covariant differentiation

[Definition] Gradient. Let $f \in C^\infty(M)$. The gradient of f is the unique vector field $\text{grad } f$ satisfying

$$g(\text{grad } f, X) = df(X) = X(f)$$

for every vector field X .

In local coordinates $x^1, \dots, x^n$, write

$$g = g_{ij} dx^i \otimes dx^j, \quad \text{grad } f = A^i \partial_i.$$

Taking $X = \partial_j$ in the defining identity gives

$$g_{ij} A^i = \partial_j f.$$

Multiplying by the inverse matrix g^{kj} yields

$$A^k = g^{kj} \partial_j f.$$

Therefore the local-coordinate formula is

$$\text{grad } f = g^{ij} \frac{\partial f}{\partial x^j} \partial_i.$$

[Definition] Musical isomorphisms. The metric converts vectors into covectors and covectors into vectors.

The musical map $\flat$ is

$$\flat : TM \longrightarrow T^*M, \quad X \longmapsto X^\flat,$$

where

$$X^\flat(Y) = g(X, Y).$$

The musical map $\sharp$ is its inverse:

$$\sharp : T^*M \longrightarrow TM, \quad \alpha \longmapsto \alpha^\sharp,$$

characterized by

$$g(\alpha^\sharp, Y) = \alpha(Y)$$

for every vector field Y . Hence

$$(X^\flat)^\sharp = X, \quad (\alpha^\sharp)^\flat = \alpha.$$

In local coordinates, if

$$X = X^i \partial_i, \quad \alpha = \alpha_i dx^i,$$

then

$$X^\flat = g_{ij} X^i dx^j, \quad \alpha^\sharp = g^{ij} \alpha_j \partial_i.$$

In particular,

$$\text{grad } f = (df)^\sharp.$$

[Remark] The differential df is defined by the smooth structure alone and is independent of the Riemannian metric. By contrast, $\text{grad } f$, $\sharp$, and $\flat$ all depend on the chosen metric g .

[Definition] Hessian. We always write $\text{grad } f$ for the gradient of a function. We reserve ∇ for covariant differentiation and do not use ∇f to denote the gradient.

The Hessian of f is the covariant derivative of its gradient:

$$\text{Hess } f = \nabla (\text{grad } f).$$

Here ∇ is the full covariant derivative. Thus $\text{Hess } f$ is a type $(1, 1)$ tensor satisfying

$$(\text{Hess } f)(X) = \nabla_X \text{grad } f.$$

Using the metric, it is equivalently viewed as a tensor of type $(0, 2)$:

$$\text{Hess } f(X, Y) = g(\nabla_X \text{grad } f, Y) = (\nabla df)(X, Y).$$

Since $Y(f) = g(\text{grad } f, Y)$, metric compatibility gives

$$\begin{aligned} X(Y(f)) &= X(g(\text{grad } f, Y)) \\ &= g(\nabla_X \text{grad } f, Y) + g(\text{grad } f, \nabla_X Y). \end{aligned}$$

Rearranging yields

$$\text{Hess } f(X, Y) = X(Y(f)) - g(\nabla_X Y, \text{grad } f).$$

In local coordinates,

$$\text{Hess } f = \left(\partial_i \partial_j f - \Gamma_{ij}^k \partial_k f \right) dx^i \otimes dx^j.$$

[Definition] Divergence. For a vector field X , its divergence is the trace of the covariant derivative ∇X :

$$\text{div } X = \text{tr}(\nabla X).$$

If $e_1, \dots, e_n$ is a local orthonormal frame, then

$$\text{div } X = \sum_{i=1}^n g(\nabla_{e_i} X, e_i).$$

In local coordinates, for $X = X^i \partial_i$,

$$\operatorname{div} X = \partial_i X^i + \Gamma_{ik}^i X^k.$$

Writing $|g| = \det(g_{ij})$, this is equivalently

$$\operatorname{div} X = \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} X^i \right).$$

[Definition] Laplacian. The Laplace–Beltrami operator applied to f is defined by

$$\Delta f = \operatorname{div} (\operatorname{grad} f).$$

Equivalently, it is the metric trace of the Hessian:

$$\Delta f = \operatorname{tr}_g (\operatorname{Hess} f).$$

To see the equivalence, choose a local orthonormal frame $e_1, \dots, e_n$. By the definitions of divergence and Hessian,

$$\begin{aligned} \operatorname{div} (\operatorname{grad} f) &= \sum_{i=1}^n g(\nabla_{e_i} \operatorname{grad} f, e_i) \\ &= \sum_{i=1}^n \operatorname{Hess} f(e_i, e_i) \\ &= \operatorname{tr}_g (\operatorname{Hess} f). \end{aligned}$$

Using the local-coordinate expression for the Hessian gives

$$\Delta f = g^{ij} \left(\partial_i \partial_j f - \Gamma_{ij}^k \partial_k f \right).$$

Equivalently, since

$$\operatorname{grad} f = g^{ij} \partial_j f \partial_i,$$

the divergence formula gives

$$\Delta f = \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \partial_j f \right), \quad |g| = \det(g_{ij}).$$

The two coordinate formulas agree as well. Expanding the divergence expression gives

$$\begin{aligned} \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \partial_j f \right) &= g^{ij} \partial_i \partial_j f \\ &\quad + \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \right) \partial_j f. \end{aligned}$$

We now derive the contracted identity. Metric compatibility also holds for the inverse metric, so

$$\nabla_i g^{ij} = 0.$$

Here ∇ is the extension of covariant differentiation from vector fields to general tensor fields defined above. Since g^{-1} is a tensor of type $(2,0)$, the extended derivative acts on both contravariant indices. Expanding it gives

$$0 = \partial_i g^{ij} + \Gamma_{ik}^i g^{kj} + \Gamma_{ik}^j g^{ik}.$$

The derivative of a determinant satisfies

$$\partial_k \log \sqrt{|g|} = \frac{1}{2} g^{ab} \partial_k g_{ab}.$$

On the other hand, contracting the Christoffel formula gives

$$\Gamma_{ik}^i = \frac{1}{2} g^{ab} \partial_k g_{ab}.$$

Therefore

$$\Gamma_{ik}^i = \partial_k \log \sqrt{|g|} = \frac{1}{\sqrt{|g|}} \partial_k \sqrt{|g|}.$$

Using the product rule,

$$\begin{aligned} \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \right) &= \partial_i g^{ij} + g^{ij} \partial_i \log \sqrt{|g|} \\ &= \partial_i g^{ij} + g^{ij} \Gamma_{ki}^k. \end{aligned}$$

After renaming dummy indices, the first two terms on the right are exactly the first two terms in the expanded equation $\nabla_i g^{ij} = 0$. Hence

$$\frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \right) = -g^{ik} \Gamma_{ik}^j = -g^{ab} \Gamma_{ab}^j.$$

Substitution yields

$$\begin{aligned} \Delta f &= g^{ij} \partial_i \partial_j f - g^{ab} \Gamma_{ab}^j \partial_j f \\ &= g^{ij} \left(\partial_i \partial_j f - \Gamma_{ij}^k \partial_k f \right), \end{aligned}$$

where the indices in the second term were renamed. Hence the divergence and Hessian definitions give exactly the same operator.

11 Curvature tensor

[Definition] The curvature tensor measures the failure of second covariant derivatives to commute. For vector fields X, Y, Z , define

$$\mathcal{R}(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Equivalently,

$$\mathcal{R}(X, Y)Z = \left([\nabla_X, \nabla_Y] - \nabla_{[X, Y]} \right) Z.$$

Although the formula contains derivatives of the vector fields, all derivative terms involving scalar coefficients cancel. Consequently, $\mathcal{R}(X, Y)Z$ is $C^\infty(M)$ -linear in X , Y , and Z . Thus

$$\mathcal{R} \in \Gamma(TM \otimes \otimes^3 T^*M)$$

is a tensor of type $(1, 3)$. At a point p , the value $\mathcal{R}_p(X, Y)Z$ depends only on X_p, Y_p, Z_p .

Using the metric, lower the output index and define the type $(0, 4)$ curvature tensor by

$$\mathcal{R}(X, Y, Z, W) = g(\mathcal{R}(X, Y)Z, W).$$

[Lemma] Algebraic symmetries. With the sign convention above, the curvature tensor satisfies

$$\mathcal{R}(X, Y)Z = -\mathcal{R}(Y, X)Z,$$

and the first Bianchi identity

$$\mathcal{R}(X, Y)Z + \mathcal{R}(Y, Z)X + \mathcal{R}(Z, X)Y = 0.$$

The type $(0, 4)$ tensor also satisfies

$$\mathcal{R}(X, Y, Z, W) = -\mathcal{R}(X, Y, W, Z)$$

and the pair symmetry

$$\mathcal{R}(X, Y, Z, W) = \mathcal{R}(Z, W, X, Y).$$

Together with skew-symmetry in the first two variables, these imply all the usual curvature symmetries.

The differential symmetry is the second Bianchi identity:

$$(\nabla_X \mathcal{R})(Y, Z) + (\nabla_Y \mathcal{R})(Z, X) + (\nabla_Z \mathcal{R})(X, Y) = 0.$$

Here each term is an endomorphism of TM . Equivalently, the identity may be evaluated on an additional vector field.

[Definition] Sectional curvature. Let $p \in M$ and let $u, v \in T_p M$ be linearly independent. The sectional curvature of the plane

$$\sigma = \text{span}\{u, v\}$$

is

$$\text{Sec}(u, v) = \frac{\mathcal{R}(u, v, v, u)}{\|u \wedge v\|^2}.$$

Since

$$\|u \wedge v\|^2 = \|u\|^2 \|v\|^2 - g(u, v)^2,$$

we obtain

$$\text{Sec}(u, v) = \frac{g(\mathcal{R}(u, v)v, u)}{\|u\|^2\|v\|^2 - g(u, v)^2}.$$

This value depends only on the plane σ , not on the chosen basis u, v . We may therefore write $\text{Sec}(\sigma)$. The sectional curvatures of all two-planes determine the full curvature tensor by polarization.

[Definition] Ricci curvature. For $u, v \in T_p M$, define

$$\text{Ric}(u, v) = \text{tr}(X \mapsto \mathcal{R}(X, u)v).$$

If $e_1, \dots, e_n$ is an orthonormal basis of $T_p M$, then

$$\text{Ric}(u, v) = \sum_{i=1}^n g(\mathcal{R}(e_i, u)v, e_i).$$

The curvature symmetries imply that Ric is a symmetric tensor of type $(0, 2)$.

[Definition] Scalar curvature. The scalar curvature is the metric trace of the Ricci tensor:

$$S = \text{tr}_g \text{Ric} = \sum_{i=1}^n \text{Ric}(e_i, e_i).$$

For an orthonormal basis,

$$S = \sum_{i,j=1}^n \mathcal{R}(e_j, e_i, e_i, e_j) = 2 \sum_{i < j} \text{Sec}(e_i, e_j).$$

Thus scalar curvature is twice the sum of the sectional curvatures of the coordinate two-planes determined by an orthonormal basis.

[Definition] Curvature operator. The space $\Lambda^2 T_p M$ is the space of bivectors. Define a symmetric bilinear form on bivectors by

$$\mathcal{R}(X \wedge Y, Z \wedge W) = \mathcal{R}(X, Y, Z, W).$$

The curvature symmetries ensure that this is well-defined. The metric on $T_p M$ induces an inner product on $\Lambda^2 T_p M$, so $\mathcal{R}$ may equivalently be viewed as the self-adjoint operator

$$\mathcal{R} : \Lambda^2 T_p M \longrightarrow \Lambda^2 T_p M$$

characterized by

$$\langle \mathcal{R}(X \wedge Y), Z \wedge W \rangle = \mathcal{R}(X, Y, Z, W).$$

Regarding $\mathcal{R}$ as the symmetric bilinear form on $\Lambda^2 T_p M$ defined above, it is positive if

$$\mathcal{R}(\omega, \omega) > 0$$

for every nonzero $\omega \in \Lambda^2 T_p M$. Positive sectional curvature requires positivity only on decomposable bivectors $\omega = u \wedge v$. Therefore a positive curvature operator implies positive sectional curvature, while the converse need not hold.

12 Some important problems

Curvature is a local quantity, but strong curvature conditions often constrain the global geometry and topology of a manifold. Several central problems ask which manifolds admit metrics satisfying a prescribed curvature condition and how rigid such metrics must be.

Existence of constant-curvature metrics

[Definition] Constant sectional curvature.

A Riemannian manifold has constant sectional curvature k if

$$\text{Sec}(\sigma) = k$$

for every tangent two-plane σ . The complete simply connected models are

$$S^n \quad k > 0, \quad \mathbb{R}^n \quad k = 0, \quad \mathbb{H}^n \quad k < 0,$$

after rescaling the metrics to obtain the desired value of k .

[Definition] A **space form** is a complete, connected Riemannian manifold with constant sectional curvature. Every space form is a quotient

$$M = \widetilde{M}/\Gamma,$$

where $\widetilde{M}$ is S^n , $\mathbb{R}^n$, or $\mathbb{H}^n$ according as the curvature is positive, zero, or negative, and Γ is a discrete group of isometries acting freely and properly discontinuously.

[Definition] A **spherical space form** is a space form of positive sectional curvature. After normalizing the curvature to 1, it has the form

$$M = S^n/\Gamma,$$

where Γ is a finite group of isometries acting freely on S^n . Examples include S^n itself and real projective space $\mathbb{RP}^n = S^n/\{\pm 1\}$.

In dimension two, closed constant-curvature surfaces are organized by their topology. The sphere and its quotients carry positive curvature, the torus and Klein bottle carry flat metrics, and closed surfaces with negative Euler characteristic carry hyperbolic metrics. Thus, for orientable closed surfaces, the sign is respectively positive, zero, or negative for genus 0, 1, or at least 2.

In dimensions at least three, asking which manifolds admit a constant-sectional-curvature metric leads to the study of spherical, Euclidean, and hyperbolic space forms.

[Remark] Constant scalar curvature. Given a compact Riemannian manifold (M, g) , one may ask whether a conformally related metric

$$\tilde{g} = e^{2f} g$$

has constant scalar curvature. This is the Yamabe problem. The answer is yes: every conformal class on a compact manifold contains a constant-scalar-curvature metric. In particular, this settles the problem in dimension three. The last unresolved cases of the Yamabe problem were solved by Richard Schoen. The solution need not be unique.

[Definition] Einstein metrics. A much stronger condition is the Einstein equation

$$\text{Ric} = \lambda g$$

for some constant $\lambda \in \mathbb{R}$. Such a metric is called an Einstein metric. Taking the trace with respect to g gives

$$S = n\lambda.$$

In fact, when $n \geq 3$, it is unnecessary to assume at first that the proportionality factor is constant. Suppose

$$\text{Ric} = fg$$

for some smooth function f . Taking the trace gives $S = nf$. The contracted second Bianchi identity therefore yields

$$\text{div}(\text{Ric}) = \frac{1}{2}dS = \frac{n}{2}df.$$

On the other hand, metric compatibility $\nabla g = 0$ gives

$$\text{div}(\text{Ric}) = \text{div}(fg) = df.$$

Comparing the two expressions, we obtain

$$\left(1 - \frac{n}{2}\right) df = 0.$$

Hence $df = 0$ for $n \geq 3$, so f is constant on every connected component of M .

Not every compact manifold admits an Einstein metric; there are topological and geometric obstructions. In dimension three, every Einstein metric has constant sectional curvature. Dimension four is substantially richer, and the classification of compact Einstein four-manifolds remains incomplete. In the compact Kähler setting, the sign of the first Chern class separates the negative, zero, and positive cases; the positive case is the most delicate.

Curvature and topology

Some fundamental rigidity results illustrate how curvature controls global topology.

- A complete noncompact manifold with $\text{Sec} > 0$ is diffeomorphic to $\mathbb{R}^n$.
- A complete noncompact manifold with $\text{Sec} \geq 0$ contains a compact totally geodesic soul and is diffeomorphic to the normal bundle of that soul.
- A compact three-manifold with positive Ricci curvature evolves under a normalized Ricci flow toward a constant-positive-curvature metric; in particular, it is diffeomorphic to a spherical space form.
- A compact manifold of dimension at least four with positive curvature operator evolves under normalized Ricci flow toward a constant-positive-curvature metric and is diffeomorphic to a spherical space form.
- A compact simply connected manifold whose positive sectional curvature is pointwise strictly quarter-pinched is diffeomorphic to a sphere. Here quarter-pinched means that, at each point,

$$\frac{\text{Sec}_{\max}}{\text{Sec}_{\min}} \in (1, 4).$$

The classification of compact manifolds admitting positive sectional curvature is still far from complete. Even in dimension four, the known simply connected examples are very restricted, and it remains an important problem to determine whether further examples exist.

The Hopf conjecture predicts that $S^2 \times S^2$ admits no Riemannian metric with strictly positive sectional curvature. More generally, it is conjectured that an even-dimensional compact manifold with positive sectional curvature has positive Euler characteristic.

Isometric embedding

[Definition] Another basic problem is to realize an abstract Riemannian manifold as a submanifold of Euclidean space. An isometric embedding

$$\psi : (M, g) \longrightarrow (\mathbb{R}^N, g_{\text{Euc}})$$

is a smooth embedding satisfying

$$\psi^* g_{\text{Euc}} = g.$$

13 Riemannian submanifolds

[Definition] Let

$$\psi : (M, g) \longrightarrow (\overline{M}, \overline{g})$$

be an isometric embedding, so

$$\psi^* \overline{g} = g.$$

For $p \in M$, the differential

$$d\psi_p : T_p M \longrightarrow T_{\psi(p)} \overline{M}$$

is an isometric linear injection. From now on, we identify M with $\psi(M)$ and identify $u \in T_p M$ with $d\psi_p(u) \in T_p \overline{M}$.

Tangent and normal spaces

[Definition] The normal space of M at p is

$$V_p M = (T_p M)^\perp \subset T_p \overline{M}.$$

Thus the ambient tangent space splits orthogonally as

$$T_p \overline{M} = T_p M \oplus V_p M.$$

The spaces $V_p M$ form the normal bundle VM . Because TM and the restricted ambient tangent bundle $T\overline{M}|_M$ are smooth vector bundles and $\overline{g}$ is smooth, the orthogonal complements $V_p M = (T_p M)^\perp$ vary smoothly with p . Hence VM inherits a natural smooth vector-bundle structure, and

$$T\overline{M}|_M = TM \oplus VM$$

is a smooth orthogonal splitting. Every $W \in T_p \overline{M}$ therefore has a unique decomposition

$$W = W^\top + W^\perp, \quad W^\top \in T_p M, \quad W^\perp \in V_p M.$$

Induced connection and second fundamental form

Let $\overline{\nabla}$ be the Levi-Civita connection of $(\overline{M}, \overline{g})$.

[Theorem] If $X, Y \in \Gamma(TM)$ are regarded as tangent vector fields along M , then

$$\nabla_X Y = \left(\overline{\nabla}_X Y \right)^\top.$$

Here arbitrary local ambient extensions may be used on the right-hand side. The formula is independent of those extensions and defines the Levi-Civita connection ∇ of the induced metric g on M .

[Proof sketch] Define

$$D_X Y = \left(\overline{\nabla}_X Y \right)^\top.$$

The locality and connection properties of $\overline{\nabla}$, together with linearity of the orthogonal projection, show that D is a well-defined connection on TM . It is torsion-free because

$$\begin{aligned}
D_X Y - D_Y X &= \left(\bar{\nabla}_X Y - \bar{\nabla}_Y X \right)^\top \\
&= [X, Y]^\top \\
&= [X, Y].
\end{aligned}$$

It is also compatible with the induced metric. Indeed,

$$\begin{aligned}
Xg(Y, Z) &= X\bar{g}(Y, Z) \\
&= \bar{g}(\bar{\nabla}_X Y, Z) + \bar{g}(Y, \bar{\nabla}_X Z) \\
&= g(D_X Y, Z) + g(Y, D_X Z).
\end{aligned}$$

Thus D is torsion-free and metric-compatible. By uniqueness of the Levi-Civita connection, $D = \nabla$.

[Definition] The normal component defines the second fundamental form

$$\Pi_p : T_p M \times T_p M \longrightarrow V_p M,$$

$$\Pi_p(u, v) = \left(\bar{\nabla}_X Y \right)_p^\perp, \quad X_p = u, \quad Y_p = v.$$

This value is independent of the chosen tangent and ambient extensions. Equivalently, for $X, Y \in \Gamma(TM)$,

$$\Pi(X, Y) = \left(\bar{\nabla}_X Y \right)^\perp.$$

The tangential and normal components together give the Gauss formula

$$\bar{\nabla}_X Y = \nabla_X Y + \Pi(X, Y).$$

The second fundamental form is bilinear and symmetric. Indeed, since the ambient Levi-Civita connection is torsion-free,

$$\begin{aligned}
\Pi(X, Y) - \Pi(Y, X) &= \left(\bar{\nabla}_X Y - \bar{\nabla}_Y X \right)^\perp \\
&= [X, Y]^\perp \\
&= 0,
\end{aligned}$$

because the bracket of two tangent vector fields is tangent to M .

Shape operator

[Definition] For a normal vector $\xi \in V_p M$, choose a local section $\tilde{\xi} \in \Gamma(VM)$ with $\tilde{\xi}_p = \xi$. The shape operator in the normal direction ξ is

$$S_\xi : T_p M \longrightarrow T_p M, \quad S_\xi(u) = - \left(\bar{\nabla}_X \tilde{\xi} \right)_p^\top,$$

where $X \in \Gamma(TM)$ satisfies $X_p = u$. This definition is independent of the extensions of u and ξ . The shape operator is characterized by

$$g(S_\xi u, v) = \bar{g}(\Pi(u, v), \xi).$$

To verify this identity, extend u, v to tangent fields X, Y . Since $\bar{g}(Y, \tilde{\xi}) = 0$, metric compatibility gives

$$\begin{aligned} 0 &= X \bar{g}(Y, \tilde{\xi}) \\ &= \bar{g}(\nabla_X Y, \tilde{\xi}) + \bar{g}(Y, \nabla_X \tilde{\xi}). \end{aligned}$$

Evaluating at p yields

$$\bar{g}(\Pi(u, v), \xi) = -\bar{g}\left(v, \left(\nabla_X \tilde{\xi}\right)_p^\top\right) = g(S_\xi u, v).$$

Since Π is symmetric, S_ξ is self-adjoint with respect to g .

Day 3

14 Principal curvatures and the Gauss equations

[Definition] Fix $p \in M$ and a unit normal vector $\xi \in V_p M$. The scalar second fundamental form in the direction ξ is

$$\Pi_\xi(u, v) = \bar{g}(\Pi(u, v), \xi) = g(S_\xi u, v).$$

Because S_ξ is self-adjoint, it admits a g -orthonormal eigenbasis. Its eigenvalues are called the principal curvatures of M at p in the normal direction ξ . Equivalently, if

$$S_\xi e_i = \kappa_i e_i,$$

then

$$\Pi_\xi(e_i, e_j) = \kappa_i \delta_{ij}.$$

[Remark] Replacing ξ by $-\xi$ changes every principal curvature to its negative. Thus the signs of the principal curvatures depend on the choice of normal, whereas the Gauss equation below is independent of that choice.

Let $\mathcal{R}$ and $\bar{\mathcal{R}}$ denote the curvature tensors of (M, g) and $(\bar{M}, \bar{g})$, respectively.

[Theorem] Gauss equations. For $u, v, w, z \in T_p M$,

$$\mathcal{R}(u, v)w = (\bar{\mathcal{R}}(u, v)w)^\top + S_{\Pi(v, w)}u - S_{\Pi(u, w)}v.$$

Equivalently, after pairing with z ,

$$\begin{aligned}
g(\mathcal{R}(u, v)w, z) &= \bar{g}(\overline{\mathcal{R}}(u, v)w, z) \\
&\quad + \bar{g}(\Pi(u, z), \Pi(v, w)) \\
&\quad - \bar{g}(\Pi(u, w), \Pi(v, z)).
\end{aligned}$$

Consequently, for linearly independent $u, v \in T_p M$,

$$\text{Sec}_M(u, v) = \text{Sec}_{\overline{M}}(u, v) + \frac{\bar{g}(\Pi(u, u), \Pi(v, v)) - \|\Pi(u, v)\|^2}{\|u\|^2\|v\|^2 - g(u, v)^2}.$$

Here $\text{Sec}_{\overline{M}}(u, v)$ means the ambient sectional curvature of the tangent two-plane spanned by u and v .

[Proof] Extend u, v, w locally to tangent vector fields X, Y, Z . The Gauss formula and the decomposition of the ambient derivative of a normal field give

$$\overline{\nabla}_X Z = \nabla_X Z + \Pi(X, Z),$$

and

$$(\overline{\nabla}_X \xi)^\top = -S_\xi X.$$

Therefore,

$$\begin{aligned}
(\overline{\nabla}_X \overline{\nabla}_Y Z)^\top &= (\overline{\nabla}_X (\nabla_Y Z + \Pi(Y, Z)))^\top \\
&= \nabla_X \nabla_Y Z - S_{\Pi(Y, Z)} X.
\end{aligned}$$

Interchanging X and Y gives

$$(\overline{\nabla}_Y \overline{\nabla}_X Z)^\top = \nabla_Y \nabla_X Z - S_{\Pi(X, Z)} Y,$$

while

$$(\overline{\nabla}_{[X, Y]} Z)^\top = \nabla_{[X, Y]} Z.$$

Substitution into the definition of curvature yields

$$(\overline{\mathcal{R}}(X, Y)Z)^\top = \mathcal{R}(X, Y)Z - S_{\Pi(Y, Z)} X + S_{\Pi(X, Z)} Y,$$

which is the first formula. Pairing it with W and using

$$g(S_\xi X, W) = \bar{g}(\Pi(X, W), \xi)$$

gives the scalar Gauss equation. Taking $w = v$ and $z = u$, then dividing by the squared area

$$\|u \wedge v\|^2 = \|u\|^2\|v\|^2 - g(u, v)^2,$$

gives the sectional-curvature formula.

15 The round sphere and its geodesics

[Example] Consider the round sphere

$$S^n(r) = \{x \in \mathbb{R}^{n+1} : \|x\|^2 = r^2\}$$

with the metric induced from Euclidean space. Its outward unit normal is

$$\xi_p = \frac{p}{r}.$$

If $w \in T_p S^n(r)$, then

$$\bar{\nabla}_w \xi = \frac{1}{r} w,$$

so our sign convention for the shape operator gives

$$S_\xi w = -\frac{1}{r} w.$$

Thus all principal curvatures relative to the outward normal are $-1/r$, and

$$\text{II}(u, v) = -\frac{1}{r} g(u, v) \xi.$$

Since the ambient Euclidean curvature vanishes, the Gauss equation gives

$$g(\mathcal{R}(u, v)v, u) = \frac{1}{r^2} (\|u\|^2 \|v\|^2 - g(u, v)^2),$$

and hence

$$\text{Sec}_{S^n(r)} = \frac{1}{r^2}.$$

Geodesics on the sphere. The unit-speed geodesics of $S^n(r)$ are precisely the great circles parametrized by arc length. More explicitly, if

$$\gamma(0) = p, \quad \dot{\gamma}(0) = v, \quad \|p\| = r, \quad \|v\| = 1, \quad \langle p, v \rangle = 0,$$

then

$$\gamma(t) = \cos\left(\frac{t}{r}\right) p + r \sin\left(\frac{t}{r}\right) v.$$

[Proof] The curve γ is a geodesic exactly when

$$(\ddot{\gamma})^\top = 0.$$

The normal space to the sphere at $\gamma(t)$ is spanned by $\gamma(t)$, so

$$\ddot{\gamma} = f(t)\gamma$$

for some function f . Since $\|\gamma\|^2 = r^2$,

$$\langle \gamma, \dot{\gamma} \rangle = 0$$

and

$$\langle \gamma, \ddot{\gamma} \rangle + \|\dot{\gamma}\|^2 = 0.$$

Metric compatibility shows that a geodesic has constant speed. For unit speed,

$$f(t)r^2 = \langle \gamma, \ddot{\gamma} \rangle = -1,$$

so

$$\ddot{\gamma} = -\frac{1}{r^2}\gamma.$$

Solving this linear equation with the prescribed initial data gives the stated great-circle formula.

16 Normal connection and the Codazzi equation

The ambient connection also induces a connection on the normal bundle.

[Definition] For $X \in \Gamma(TM)$ and $\xi \in \Gamma(VM)$, define the normal connection by

$$\nabla_X^\perp \xi = \left(\overline{\nabla}_X \xi \right)^\perp.$$

Together with the definition of the shape operator, this gives the Weingarten formula

$$\overline{\nabla}_X \xi = -S_\xi X + \nabla_X^\perp \xi.$$

Thus $\nabla^\perp$ is the normal projection of the ambient connection, just as ∇ is its tangential projection along tangent fields.

[Definition] The covariant derivative of the second fundamental form is defined by

$$\begin{aligned} (\nabla_X \Pi)(Y, Z) &:= \nabla_X^\perp (\Pi(Y, Z)) \\ &\quad - \Pi(\nabla_X Y, Z) - \Pi(Y, \nabla_X Z). \end{aligned}$$

[Theorem] Codazzi equation. For $u, v, w \in T_p M$,

$$(\overline{\mathcal{R}}(u, v)w)^\perp = (\nabla_u \Pi)(v, w) - (\nabla_v \Pi)(u, w).$$

[Proof] Choose local tangent extensions X, Y, Z . Taking the normal component in the curvature expansion gives

$$\begin{aligned} (\overline{\nabla}_X \overline{\nabla}_Y Z)^\perp &= (\overline{\nabla}_X (\nabla_Y Z + \Pi(Y, Z)))^\perp \\ &= \Pi(X, \nabla_Y Z) + \nabla_X^\perp (\Pi(Y, Z)). \end{aligned}$$

Subtracting the corresponding expression with X and Y interchanged and then subtracting

$$(\bar{\nabla}_{[X,Y]}Z)^\perp = \Pi([X,Y], Z)$$

yields

$$\begin{aligned} (\bar{\mathcal{R}}(X, Y)Z)^\perp &= \nabla_X^\perp(\Pi(Y, Z)) - \nabla_Y^\perp(\Pi(X, Z)) \\ &\quad + \Pi(X, \nabla_Y Z) - \Pi(Y, \nabla_X Z) - \Pi([X, Y], Z). \end{aligned}$$

Because the induced connection is torsion-free,

$$[X, Y] = \nabla_X Y - \nabla_Y X.$$

Regrouping the terms according to the preceding definition gives

$$(\bar{\mathcal{R}}(X, Y)Z)^\perp = (\nabla_X \Pi)(Y, Z) - (\nabla_Y \Pi)(X, Z),$$

as required.

17 Distance functions and level hypersurfaces

[Definition] Let $U \subset M$ be open. A smooth function

$$r : U \longrightarrow \mathbb{R}$$

is called a distance function on U if

$$\|\text{grad } r\| = 1$$

throughout U . We write

$$\partial_r := \text{grad } r.$$

For any $c \in \mathbb{R}$ such that

$$\Sigma_c := r^{-1}(c)$$

is nonempty, dr never vanishes on Σ_c . The regular-level-set theorem therefore shows that Σ_c is a smooth hypersurface of M . We equip it with the metric induced by the inclusion

$$\iota_c : \Sigma_c \hookrightarrow M, \quad g_c = \iota_c^* g.$$

The unit normal field to every level hypersurface is ∂_r .

[Example] On $\mathbb{R}^n \setminus \{0\}$ with the Euclidean metric,

$$r(x) = \|x\|$$

is a distance function. Its level hypersurfaces are the Euclidean spheres

$$\Sigma_c = S^{n-1}(c).$$

[Remark] If $A \subset M$ is closed, the function

$$r(x) = d(A, x)$$

is generally not smooth everywhere. It is a smooth distance function only on regions where the nearest point of A is unique and no cut-locus or focal-point singularity occurs. All calculations below are made on such a smooth region.

[Proposition] Let $u, v \in T_p \Sigma_c$. With the normal ∂_r , the scalar second fundamental form of Σ_c satisfies

$$\text{II}_{\partial_r}(u, v) = -\text{Hess } r(u, v).$$

Equivalently, its shape operator is

$$S_{\partial_r} u = -\nabla_u \partial_r.$$

[Proof] Since v is tangent to Σ_c ,

$$g(v, \partial_r) = dr(v) = 0.$$

Therefore metric compatibility gives

$$\begin{aligned} \text{II}_{\partial_r}(u, v) &= g(\nabla_u v, \partial_r) \\ &= u g(v, \partial_r) - g(v, \nabla_u \partial_r) \\ &= -g(v, \nabla_u \text{grad } r) \\ &= -\text{Hess } r(u, v). \end{aligned}$$

Moreover,

$$g(\nabla_u \partial_r, \partial_r) = \frac{1}{2} u g(\partial_r, \partial_r) = 0,$$

so $\nabla_u \partial_r$ is tangent to Σ_c . The definition of the shape operator now gives

$$S_{\partial_r} u = -(\nabla_u \partial_r)^\top = -\nabla_u \partial_r.$$

It will be convenient to reverse this sign and define the level-set endomorphism

$$\mathcal{S} := -S_{\partial_r}, \quad \mathcal{S}(X) = \nabla_X \partial_r$$

for vectors X tangent to the level hypersurfaces. Then

$$g(\mathcal{S}u, v) = \text{Hess } r(u, v).$$

[Lemma] The vector field ∂_r is geodesic:

$$\nabla_{\partial_r} \partial_r = 0.$$

Consequently, its integral curves are unit-speed geodesics orthogonal to every level hypersurface Σ_c .

[Proof] For any vector field Y , symmetry of the Hessian gives

$$\begin{aligned} g(\nabla_{\partial_r} \partial_r, Y) &= \text{Hess } r(\partial_r, Y) \\ &= \text{Hess } r(Y, \partial_r) \\ &= g(\nabla_Y \partial_r, \partial_r) \\ &= \frac{1}{2} Y \|\partial_r\|^2 \\ &= 0. \end{aligned}$$

Since this holds for every Y , the claim follows.

[Remark] Locally, the flow of ∂_r carries one level hypersurface to nearby levels. Equivalently, after choosing one level Σ_{c_0} , the normal exponential map gives coordinates

$$(t, p) \mapsto \exp_p(t\partial_r)$$

in which the level hypersurfaces are obtained by varying t .

Along a level hypersurface, define the radial curvature endomorphism by

$$\mathcal{R}_{\partial_r}(X) := \mathcal{R}(X, \partial_r)\partial_r.$$

[Theorem] Riccati equation. Along the integral curves of ∂_r , the endomorphism $\mathcal{S}$ satisfies

$$\nabla_{\partial_r} \mathcal{S} + \mathcal{S}^2 + \mathcal{R}_{\partial_r} = 0.$$

Thus, for every vector field X tangent to the level hypersurfaces,

$$(\nabla_{\partial_r} \mathcal{S})(X) + \mathcal{S}^2(X) + \mathcal{R}(X, \partial_r)\partial_r = 0.$$

[Proof] Extend X along the flow of ∂_r so that

$$[\partial_r, X] = 0.$$

Because the connection is torsion-free,

$$\nabla_{\partial_r} X = \nabla_X \partial_r = \mathcal{S}(X).$$

Using $\nabla_{\partial_r} \partial_r = 0$ and the curvature convention of these notes,

$$\begin{aligned} \mathcal{R}(\partial_r, X)\partial_r &= \nabla_{\partial_r} \nabla_X \partial_r - \nabla_X \nabla_{\partial_r} \partial_r - \nabla_{[\partial_r, X]} \partial_r \\ &= \nabla_{\partial_r} (\mathcal{S}(X)). \end{aligned}$$

On the other hand,

$$\begin{aligned}
(\nabla_{\partial_r} \mathcal{S})(X) &= \nabla_{\partial_r}(\mathcal{S}(X)) - \mathcal{S}(\nabla_{\partial_r} X) \\
&= \mathcal{R}(\partial_r, X)\partial_r - \mathcal{S}^2(X) \\
&= -\mathcal{R}(X, \partial_r)\partial_r - \mathcal{S}^2(X).
\end{aligned}$$

Rearranging proves the Riccati equation.

[Proposition] Curvature of a rotationally symmetric warped product. Let

$$M = I \times S^{n-1}, \quad g = dr^2 \oplus f(r)^2 g_{S^{n-1}},$$

where $I \subset \mathbb{R}$ is an interval and $f : I \rightarrow (0, \infty)$ is smooth. For a unit vector X tangent to the spherical slice and an orthonormal pair X, Y tangent to that slice,

$$\text{Sec}(X, \partial_r) = -\frac{f''}{f},$$

and

$$\text{Sec}(X, Y) = \frac{1 - (f')^2}{f^2}.$$

Consequently,

$$\text{Ric}(\partial_r, \partial_r) = -(n-1)\frac{f''}{f},$$

while for a unit vector X tangent to a slice,

$$\text{Ric}(X, X) = -\frac{f''}{f} + (n-2)\frac{1 - (f')^2}{f^2}.$$

The scalar curvature is

$$S = -2(n-1)\frac{f''}{f} + (n-1)(n-2)\frac{1 - (f')^2}{f^2}.$$

[Proof] The function r is a distance function, and its level hypersurfaces are the slices

$$\Sigma_{r_0} = \{r_0\} \times S^{n-1}.$$

Their induced metric is

$$g_{r_0} = f(r_0)^2 g_{S^{n-1}},$$

so each slice has intrinsic sectional curvature

$$\text{Sec}_{\Sigma_{r_0}} = \frac{1}{f(r_0)^2}.$$

For vectors X, Y tangent to the slices, the standard variation formula for the induced metric gives

$$2 \text{Hess } r(X, Y) = \mathcal{L}_{\partial_r} g(X, Y).$$

Since

$$\mathcal{L}_{\partial_r} g = 2f(r)f'(r)g_{S^{n-1}} = 2\frac{f'(r)}{f(r)}g$$

on the tangent spaces of the slices, we obtain

$$\text{Hess } r(X, Y) = \frac{f'}{f}g(X, Y).$$

Therefore the level-set endomorphism is

$$\mathcal{S}(X) = \nabla_X \partial_r = \frac{f'}{f}X.$$

Apply the Riccati equation to X . Because $\mathcal{S} = (f'/f)\text{Id}$,

$$\left(\nabla_{\partial_r} \mathcal{S} + \mathcal{S}^2\right)(X) = \frac{f''}{f}X.$$

It follows that

$$\mathcal{R}(X, \partial_r)\partial_r = -\frac{f''}{f}X,$$

which proves the radial sectional-curvature formula.

For a tangential orthonormal pair X, Y , the shape operator in the normal direction ∂_r is

$$S_{\partial_r} = -\mathcal{S} = -\frac{f'}{f}\text{Id}.$$

The Gauss equation for the hypersurface $\Sigma_{r_0} \subset M$ gives

$$\text{Sec}_{\Sigma_{r_0}}(X, Y) = \text{Sec}_M(X, Y) + \left(\frac{f'}{f}\right)^2.$$

Since $\text{Sec}_{\Sigma_{r_0}} = 1/f^2$,

$$\text{Sec}_M(X, Y) = \frac{1 - (f')^2}{f^2}.$$

Tracing the two types of sectional curvature gives the stated Ricci and scalar-curvature formulas.

[Corollary] Constant-sectional-curvature models. The warped product above has constant sectional curvature K precisely when

$$-\frac{f''}{f} = K, \quad \frac{1 - (f')^2}{f^2} = K.$$

Equivalently,

$$f'' + Kf = 0, \quad (f')^2 + Kf^2 = 1.$$

To close one end of the warped product smoothly by adding a pole at $r = 0$, impose

$$f(0) = 0, \quad f'(0) = 1.$$

The solutions are

$$f(r) = \text{sn}_K(r) := \begin{cases} \frac{1}{\sqrt{K}} \sin(\sqrt{K} r), & K > 0, \\ r, & K = 0, \\ \frac{1}{\sqrt{-K}} \sinh(\sqrt{-K} r), & K < 0. \end{cases}$$

For $K = 0$, adding the pole produces Euclidean space. For $K < 0$, it produces the complete simply connected hyperbolic space of curvature K . For $K > 0$, take

$$0 < r < \frac{\pi}{\sqrt{K}}$$

and add a pole at each endpoint; the resulting compact manifold is the round sphere of sectional curvature K .

18 Geodesics and the exponential map

The exponential map converts tangent vectors into points of the manifold by following geodesics. It therefore provides canonical local coordinates and allows geometric comparison problems to be transferred to tangent spaces.

[Definition] A C^2 curve $\gamma : (a, b) \rightarrow M$ is a geodesic if

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0.$$

If $\gamma(t) = (x^1(t), \dots, x^n(t))$ in local coordinates, this is the system

$$\frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k(\gamma(t)) \frac{dx^i}{dt} \frac{dx^j}{dt} = 0, \quad k = 1, \dots, n.$$

Thus the geodesic equation is a nonlinear second-order ordinary differential equation.

[Theorem] Local existence and uniqueness of geodesics. For every $p \in M$ and $v \in T_p M$, there is a unique maximal geodesic $\gamma_v : I_v \rightarrow M$ satisfying

$$\gamma_v(0) = p, \quad \dot{\gamma}_v(0) = v.$$

The interval I_v is open and contains 0, and the solution depends smoothly on the initial data (p, v) . In particular, the geodesic exists for all sufficiently small times.

[Remark] If $K \subset TM$ is compact, smooth dependence and the ODE existence theorem give a number $\tau > 0$ such that every geodesic with initial data in K is defined for $|t| < \tau$. Thus on a compact Riemannian manifold the existence time can be chosen uniformly for all unit initial vectors. Equivalently, if we insist on keeping the time interval fixed up to $t = 1$, then we must shrink the allowed initial velocities: by the scaling property of the geodesic equation, sufficiently small vectors in $T_p M$ generate geodesics that remain defined until time 1.

[Definition] For $p \in M$, choose $\delta > 0$ sufficiently small that $\gamma_v(1)$ is defined for every

$$v \in B_\delta(0_p) \subset T_p M.$$

The local exponential map at p is

$$\exp_p : B_\delta(0_p) \rightarrow M, \quad \exp_p(v) := \gamma_v(1).$$

The scaling property of the geodesic equation gives

$$\gamma_v(t) = \exp_p(tv)$$

whenever both sides are defined. Equivalently, if u is a unit vector and $v = ru$, then

$$\exp_p(v) = \gamma_u(r).$$

Since $T_p M$ is itself a vector space, its tangent space at the origin is canonically identified with $T_p M$:

$$T_{0_p}(T_p M) \cong T_p M, \quad u \longleftrightarrow \left. \frac{d}{dt} \right|_{t=0} (tu).$$

Under this identification, the differential of the exponential map is an endomorphism of $T_p M$.

[Proposition] The differential of the exponential map at the origin is the identity:

$$(d \exp_p)_{0_p} = \text{Id}_{T_p M}.$$

[Proof] For $u \in T_p M$, consider the curve $c(t) = tu$ in $T_p M$. Then

$$\exp_p(c(t)) = \exp_p(tu) = \gamma_u(t).$$

Therefore

$$\begin{aligned} (d \exp_p)_{0_p}(u) &= \left. \frac{d}{dt} \right|_{t=0} \exp_p(tu) \\ &= \dot{\gamma}_u(0) \\ &= u. \end{aligned}$$

[Corollary] There is an $\varepsilon > 0$ such that

$$\exp_p : B_\varepsilon(0_p) \subset T_p M \longrightarrow B$$

is a diffeomorphism onto an open neighbourhood B of p .

[Proof] Apply the inverse function theorem to $\exp_p$ at 0_p .

Day 4

19 The differential of the exponential map and Jacobi fields

We now study the differential of the exponential map away from the origin. Because $T_p M$ is a vector space, translation gives a canonical identification

$$T_u(T_p M) \cong T_p M$$

for every $u \in T_p M$. Thus, whenever $\exp_p$ is defined near u ,

$$(d\exp_p)_u : T_u(T_p M) \longrightarrow T_{\exp_p(u)} M$$

may be regarded as a linear map from $T_p M$ to $T_{\exp_p(u)} M$.

Fix $u \in B_\delta(0_p)$ and $\xi \in T_u(T_p M) \cong T_p M$. Consider the line

$$\ell(s) = u + s\xi$$

in $T_p M$ and the two-parameter map

$$\alpha(t, s) = \exp_p(t(u + s\xi)),$$

defined for s sufficiently small and for the relevant values of t . For each fixed s , the curve

$$c_s(t) := \alpha(t, s)$$

is the geodesic starting at p with initial velocity

$$c_s(0) = p, \quad \dot{c}_s(0) = u + s\xi.$$

[Definition] The vector field along $\gamma_u(t) = \alpha(t, 0)$ given by

$$Y(t) := \left. \frac{\partial \alpha}{\partial s}(t, s) \right|_{s=0}$$

is called the variational vector field of the variation α .

Differentiating the defining formula for α with respect to s gives

$$\begin{aligned} Y(t) &= (d \exp_p)_{tu} \left(\left. \frac{\partial}{\partial s} \right|_{s=0} t(u + s\xi) \right) \\ &= (d \exp_p)_{tu}(t\xi). \end{aligned}$$

In particular,

$$Y(1) = (d \exp_p)_u(\xi).$$

Hence the differential of $\exp_p$ can be studied by finding the differential equation satisfied by Y .

[Definition] Let

$$\alpha : (a, b) \times (-\varepsilon, \varepsilon) \rightarrow M$$

be a C^2 map. For each fixed s , set $c_s(t) = \alpha(t, s)$. The family $\{c_s\}$ is called a variation of the curve c_0 . It is a variation through geodesics if every c_s is a geodesic.

Write

$$T := \frac{\partial \alpha}{\partial t}, \quad S := \frac{\partial \alpha}{\partial s}.$$

These are vector fields along the image of α . Since the coordinate vector fields ∂_t and ∂_s commute,

$$[T, S] = 0.$$

Because the Levi-Civita connection is torsion-free,

$$\nabla_T S = \nabla_S T.$$

[Theorem] Jacobi equation. The variational vector field Y of a variation through geodesics satisfies

$$\frac{D^2 Y}{dt^2} + \mathcal{R}(Y, \dot{\gamma})\dot{\gamma} = 0,$$

where $\gamma = c_0$ and $D/dt = \nabla_{\dot{\gamma}}$ is covariant differentiation along γ .

[Proof] Since every $t \mapsto \alpha(t, s)$ is a geodesic,

$$\nabla_T T = 0.$$

Using $[T, S] = 0$, the curvature convention of these notes gives

$$\begin{aligned} \mathcal{R}(T, S)T &= \nabla_T \nabla_S T - \nabla_S \nabla_T T - \nabla_{[T, S]} T \\ &= \nabla_T \nabla_S T. \end{aligned}$$

Torsion-freeness gives $\nabla_S T = \nabla_T S$, so along $s = 0$,

$$\begin{aligned}
\frac{D^2 Y}{dt^2} &= \nabla_T \nabla_T S \\
&= \nabla_T \nabla_S T \\
&= \mathcal{R}(T, S)T \\
&= -\mathcal{R}(S, T)T.
\end{aligned}$$

Restricting to $s = 0$, where $T = \dot{\gamma}$ and $S = Y$, yields

$$\frac{D^2 Y}{dt^2} + \mathcal{R}(Y, \dot{\gamma})\dot{\gamma} = 0.$$

[Definition] A vector field Y along a geodesic γ is called a Jacobi field if it satisfies the Jacobi equation

$$\frac{D^2 Y}{dt^2} + \mathcal{R}(Y, \dot{\gamma})\dot{\gamma} = 0.$$

[Proposition] Let γ be a geodesic in an n -dimensional Riemannian manifold. For any vectors $A, B \in T_{\gamma(0)}M$, there exists a unique Jacobi field Y along γ satisfying

$$Y(0) = A, \quad \frac{DY}{dt}(0) = B.$$

Consequently, the vector space of Jacobi fields along γ has dimension $2n$.

[Remark] In a parallel frame along γ , the Jacobi equation becomes a linear system of second-order ordinary differential equations with smooth coefficients. The existence and uniqueness statement, as well as the dimension count, therefore follow from the usual theory of linear ODEs.

For the radial variation above, the initial conditions are

$$Y(0) = 0, \quad \frac{DY}{dt}(0) = \xi.$$

Thus $(d \exp_p)_u(\xi)$ is obtained by solving this Jacobi equation and evaluating its solution at $t = 1$.

20 Properties of Jacobi fields and curvature expansions

Let γ_u be the geodesic with

$$\gamma_u(0) = p, \quad \dot{\gamma}_u(0) = u.$$

Denote the vector space of Jacobi fields along γ_u by $\text{Jac}(\gamma_u)$. The initial-value theorem gives a linear isomorphism

$$\text{Jac}(\gamma_u) \longrightarrow T_p M \oplus T_p M, \quad Y \longmapsto \left(Y(0), \frac{DY}{dt}(0) \right).$$

Therefore

$$\dim \text{Jac}(\gamma_u) = 2n.$$

The subspace

$$\text{Jac}_0(\gamma_u) := \{Y \in \text{Jac}(\gamma_u) : Y(0) = 0\}$$

has dimension n .

For $t \neq 0$, the relation obtained above may also be written as

$$(d \exp_p)_{tu}(\xi) = \frac{1}{t} Y_\xi(t),$$

where Y_ξ is the unique Jacobi field satisfying

$$Y_\xi(0) = 0, \quad \frac{DY_\xi}{dt}(0) = \xi.$$

[Proposition] For $t_0 \neq 0$, the differential $(d \exp_p)_{t_0 u}$ is singular if and only if there exists a nonzero Jacobi field Y along γ_u such that

$$Y(0) = 0, \quad Y(t_0) = 0.$$

[Proof] If $\xi \in \ker(d \exp_p)_{t_0 u}$, then the corresponding field Y_ξ satisfies

$$Y_\xi(t_0) = t_0(d \exp_p)_{t_0 u}(\xi) = 0.$$

Conversely, every Jacobi field vanishing at 0 is uniquely determined by

$$\xi = \frac{DY}{dt}(0),$$

so $Y(t_0) = 0$ implies $\xi \in \ker(d \exp_p)_{t_0 u}$.

[Definition] The point $q = \gamma_u(t_0)$ is conjugate to $p = \gamma_u(0)$ along γ_u if there is a nonzero Jacobi field Y satisfying

$$Y(0) = Y(t_0) = 0.$$

The multiplicity of this conjugate point is

$$\dim \ker(d \exp_p)_{t_0 u}.$$

Thus conjugate points are exactly the points at which the exponential map becomes singular along a radial geodesic.

Assume now that γ has unit speed. If Y is a Jacobi field along γ , then

$$a(t) := g(Y(t), \dot{\gamma}(t))$$

is affine in t . Indeed,

$$\begin{aligned} a''(t) &= g\left(\frac{D^2Y}{dt^2}, \dot{\gamma}\right) \\ &= -g(\mathcal{R}(Y, \dot{\gamma})\dot{\gamma}, \dot{\gamma}) \\ &= 0. \end{aligned}$$

Consequently,

$$g(Y(t), \dot{\gamma}(t)) = a + bt,$$

where

$$a = g(Y(0), \dot{\gamma}(0)), \quad b = g\left(\frac{DY}{dt}(0), \dot{\gamma}(0)\right).$$

[Definition] A Jacobi field Y is normal if

$$g(Y(t), \dot{\gamma}(t)) = 0$$

for every t .

[Proposition] Every Jacobi field along a unit-speed geodesic decomposes uniquely as

$$Y(t) = (a + bt)\dot{\gamma}(t) + Y^\perp(t),$$

where $Y^\perp$ is a normal Jacobi field.

[Example] Suppose the manifold has constant sectional curvature K . Let

$$E_1 = \dot{\gamma}, E_2, \dots, E_n$$

be an orthonormal parallel frame along a unit-speed geodesic. A normal Jacobi field

$$Y = \sum_{i=2}^n y^i E_i$$

satisfies

$$(y^i)'' + Ky^i = 0.$$

Define

$$\text{sn}_K(t) = \begin{cases} \frac{\sin(\sqrt{K}t)}{\sqrt{K}}, & K > 0, \\ t, & K = 0, \\ \frac{\sinh(\sqrt{-K}t)}{\sqrt{-K}}, & K < 0, \end{cases} \quad \text{cn}_K(t) := \text{sn}'_K(t).$$

Then

$$y^i(t) = a_i \text{cn}_K(t) + b_i \text{sn}_K(t).$$

In particular, if $Y(0) = 0$ and $DY/dt(0) = \xi \perp \dot{\gamma}(0)$, then

$$Y(t) = \operatorname{sn}_K(t) P_t \xi,$$

where P_t denotes parallel transport along γ .

We next extract the first curvature correction to a radial Jacobi field.

[Proposition] Taylor expansion of a radial Jacobi field. Let $P_t : T_p M \rightarrow T_{\gamma_u(t)} M$ denote parallel transport. If

$$Y_\xi(0) = 0, \quad \frac{DY_\xi}{dt}(0) = \xi,$$

then, as $t \rightarrow 0$,

$$Y_\xi(t) = P_t \left[t\xi - \frac{t^3}{6} \mathcal{R}_p(\xi, u)u + O(t^4) \right].$$

[Proof] The Jacobi equation gives

$$\frac{D^2 Y_\xi}{dt^2}(0) = 0.$$

Covariantly differentiating the Jacobi equation once and using $Y_\xi(0) = 0$ and $D\dot{\gamma}_u/dt = 0$ gives

$$\frac{D^3 Y_\xi}{dt^3}(0) = -\mathcal{R}_p(\xi, u)u.$$

Taylor's formula in a parallel frame now gives the result.

[Corollary] For $\xi, \eta \in T_p M$,

$$\begin{aligned} g(Y_\xi(t), Y_\eta(t)) &= t^2 g_p(\xi, \eta) \\ &\quad - \frac{t^4}{3} \mathcal{R}_p(\xi, u, u, \eta) + O(t^5). \end{aligned}$$

[Proof] Since parallel transport is an isometry, the preceding Taylor expansion gives

$$\begin{aligned} g(Y_\xi(t), Y_\eta(t)) &= g_p \left(t\xi - \frac{t^3}{6} \mathcal{R}_p(\xi, u)u + O(t^4), t\eta - \frac{t^3}{6} \mathcal{R}_p(\eta, u)u + O(t^4) \right) \\ &= t^2 g_p(\xi, \eta) \\ &\quad - \frac{t^4}{6} [g_p(\mathcal{R}_p(\xi, u)u, \eta) + g_p(\xi, \mathcal{R}_p(\eta, u)u)] + O(t^5). \end{aligned}$$

The curvature symmetries imply

$$g_p(\mathcal{R}_p(\eta, u)u, \xi) = \mathcal{R}_p(\eta, u, u, \xi) = \mathcal{R}_p(\xi, u, u, \eta).$$

Thus the two fourth-order terms are equal, and the claimed coefficient $-1/3$ follows.

Here

$$\mathcal{R}_p(\xi, u, u, \eta) = g_p(\mathcal{R}_p(\xi, u)u, \eta).$$

[Theorem] Metric expansion in normal coordinates. Choose an orthonormal basis $e_1, \dots, e_n$ of $T_p M$ and use $\exp_p$ to define normal coordinates $x^1, \dots, x^n$ near p . Then

$$g_{ij}(x) = \delta_{ij} - \frac{1}{3}\mathcal{R}_p(e_i, x, x, e_j) + O(|x|^3).$$

Equivalently, writing $x = x^k e_k$,

$$g_{ij}(x) = \delta_{ij} - \frac{1}{3}\mathcal{R}_p(e_i, e_k, e_l, e_j)x^k x^l + O(|x|^3).$$

[Proof] Put $x = tu$. The coordinate vector ∂_i at $\exp_p(tu)$ is

$$\partial_i = (d\exp_p)_{tu}(e_i) = \frac{1}{t}Y_{e_i}(t).$$

Therefore

$$g_{ij}(tu) = \frac{1}{t^2}g(Y_{e_i}(t), Y_{e_j}(t)),$$

and the preceding corollary gives the claimed expansion.

[Corollary] In normal coordinates,

$$\sqrt{\det g(x)} = 1 - \frac{1}{6}\text{Ric}_p(x, x) + O(|x|^3).$$

Consequently, if ω_n denotes the Euclidean volume of the unit ball in $\mathbb{R}^n$, then

$$\text{Vol}(B(p, r)) = \omega_n r^n \left[1 - \frac{S(p)}{6(n+2)}r^2 + O(r^4) \right]$$

for sufficiently small r .

The first correction to Euclidean volume is therefore controlled by the scalar curvature at the center.

21 The Gauss lemma

[Theorem] Gauss lemma. Let $u, \xi \in T_p M$ and suppose that $\exp_p$ is defined at tu . Then

$$g_{\gamma_u(t)}\left((d\exp_p)_{tu}(\xi), \dot{\gamma}_u(t)\right) = g_p(\xi, u).$$

Equivalently, for $v, w \in T_p M$,

$$g_{\exp_p(v)}\left((d\exp_p)_v(v), (d\exp_p)_v(w)\right) = g_p(v, w).$$

[Proof] Let Y_ξ be the radial Jacobi field satisfying

$$Y_\xi(0) = 0, \quad \frac{DY_\xi}{dt}(0) = \xi.$$

The affine-function property gives

$$g(Y_\xi(t), \dot{\gamma}_u(t)) = t g_p(\xi, u).$$

Since

$$(d \exp_p)_{tu}(\xi) = \frac{1}{t} Y_\xi(t),$$

the first identity follows. The second is obtained by taking $t = 1$.

[Corollary] The exponential map preserves lengths in the radial direction and maps vectors orthogonal to the radial direction to vectors orthogonal to the radial geodesic. In particular,

$$\xi \perp u \implies (d \exp_p)_{tu}(\xi) \perp \dot{\gamma}_u(t).$$

Thus the radial geodesics meet the small geodesic spheres centered at p orthogonally.

Day 5

22 Distance and geodesic balls

[Definition] The length of a piecewise C^1 curve $\sigma : [a, b] \rightarrow M$ is

$$L(\sigma) = \int_a^b \|\dot{\sigma}(t)\| dt.$$

The Riemannian distance is

$$d(p, q) := \inf_{\substack{\sigma(a)=p \\ \sigma(b)=q}} L(\sigma).$$

It defines a metric on each connected component of M .

[Proposition] If M is connected, the Riemannian distance satisfies

$$d(p, q) \geq 0, \quad d(p, q) = 0 \iff p = q,$$

$$d(p, q) = d(q, p), \quad d(p, r) \leq d(p, q) + d(q, r).$$

Thus (M, d) is a metric space, and the metric topology agrees with the manifold topology.

[Proof] Nonnegativity is immediate. Reversing a curve preserves its length, which gives symmetry, and concatenating curves gives the triangle inequality. In a normal neighbourhood of p , the local minimizing result below gives

$$d(p, q) = \left\| \exp_p^{-1}(q) \right\|,$$

so $d(p, q) = 0$ forces $p = q$ and metric balls form a neighbourhood basis.

[Theorem] Radial geodesics minimize locally. Choose $\rho > 0$ so that

$$\exp_p : B_\rho(0_p) \longrightarrow U$$

is a diffeomorphism. If $0 < \|v\| < \rho$, $q = \exp_p(v)$, and

$$\gamma_v(t) = \exp_p(tv), \quad 0 \leq t \leq 1,$$

then every piecewise C^1 curve $\tilde{\gamma} : [0, 1] \rightarrow U$ from p to q satisfies

$$L(\tilde{\gamma}) \geq L(\gamma_v) = \|v\|.$$

Equality holds if and only if $\tilde{\gamma}$ is an orientation-preserving reparametrization of γ_v , allowing intervals on which the reparametrization is constant.

[Proof] Lift $\tilde{\gamma}$ through the exponential chart:

$$\varphi(t) := \exp_p^{-1}(\tilde{\gamma}(t)).$$

Then $\varphi(0) = 0_p$ and $\varphi(1) = v$. Wherever $\varphi(t) \neq 0_p$, write

$$\varphi(t) = r(t)e(t), \quad r(t) = \|\varphi(t)\|, \quad \|e(t)\| = 1.$$

Thus

$$\dot{\varphi}(t) = \dot{r}(t)e(t) + r(t)\dot{e}(t), \quad \langle e(t), \dot{e}(t) \rangle = 0.$$

By the Gauss lemma, the images of the radial and angular components under $(d\exp_p)_{\varphi(t)}$ are orthogonal, and

$$\left\| (d\exp_p)_{\varphi(t)} e(t) \right\| = \|e(t)\| = 1.$$

Consequently,

$$\begin{aligned} \left\| \dot{\tilde{\gamma}}(t) \right\|^2 &= \left\| (d\exp_p)_{\varphi(t)} (\dot{r}(t)e(t) + r(t)\dot{e}(t)) \right\|^2 \\ &= |\dot{r}(t)|^2 + \left\| (d\exp_p)_{\varphi(t)} (r(t)\dot{e}(t)) \right\|^2 \geq |\dot{r}(t)|^2. \end{aligned}$$

The same inequality holds at zeros of φ by applying the argument on the components where $r > 0$ and using continuity. Therefore

$$\begin{aligned} L(\tilde{\gamma}) &\geq \int_0^1 |\dot{r}(t)| dt \\ &\geq r(1) - r(0) \\ &= \|v\| = L(\gamma_v). \end{aligned}$$

If equality holds, then r is nondecreasing and the angular term vanishes. Since $\exp_p$ is a diffeomorphism on $B_\rho(0_p)$, this forces $\dot{e} = 0$ wherever $r > 0$. Hence

$$\tilde{\gamma}(t) = \exp_p(r(t)e_0) = \gamma_v\left(\frac{r(t)}{\|v\|}\right), \quad e_0 = \frac{v}{\|v\|},$$

so $\tilde{\gamma}$ is an orientation-preserving reparametrization of γ_v . The converse is immediate.

[Corollary] After decreasing ρ if necessary, every $q = \exp_p(v)$ with $\|v\| < \rho$ satisfies

$$d(p, q) = \|v\|.$$

[Proof] Choose ρ so that the closed ball $\overline{B_\rho(0_p)}$ lies in a larger normal neighbourhood. A curve from p to q that remains in U has length at least $\|v\|$ by the theorem. If it leaves U , its initial segment reaches the geodesic sphere of radius ρ and the same argument gives length at least $\rho > \|v\|$. The radial geodesic has length $\|v\|$, so it realizes the distance.

[Corollary] For $0 < r < \rho$,

$$B(p, r) = \exp_p(B_r(0_p)),$$

and

$$\partial B(p, r) = \exp_p(\partial B_r(0_p)).$$

The boundary $\partial B(p, r)$ is a smooth geodesic sphere.

[Corollary] On the punctured normal neighbourhood $U \setminus \{p\}$, the distance function

$$r(x) = d(p, x)$$

is smooth and satisfies

$$\|\text{grad } r\| = 1.$$

If $x = \exp_p(tu)$ with $\|u\| = 1$, then

$$\text{grad } r(x) = \dot{\gamma}_u(t).$$

Hence the integral curves of $\text{grad } r$ are precisely the outward radial unit-speed geodesics.

23 Variations of length and energy

Let $\gamma : [a, b] \rightarrow M$ be a piecewise C^1 curve. Its length and energy are

$$L(\gamma) = \int_a^b \|\dot{\gamma}\| dt, \quad E(\gamma) = \frac{1}{2} \int_a^b \|\dot{\gamma}\|^2 dt.$$

[Proposition] The Cauchy–Schwarz inequality gives

$$L(\gamma)^2 \leq 2(b-a)E(\gamma).$$

Equality holds if and only if $\|\dot{\gamma}\|$ is constant. Consequently, among curves on the fixed interval $[a, b]$ with fixed endpoints, a curve minimizes energy if and only if it minimizes length and has constant speed.

[Definition] A variation of a smooth curve $\gamma : [a, b] \rightarrow M$ is a smooth map

$$\alpha : [a, b] \times (-\varepsilon, \varepsilon) \longrightarrow M$$

such that $\alpha(t, 0) = \gamma(t)$. Write

$$\gamma_s(t) = \alpha(t, s), \quad T = \frac{\partial \alpha}{\partial t}, \quad V = \frac{\partial \alpha}{\partial s} \Big|_{s=0}.$$

The variation has fixed endpoints if $\alpha(a, s) = \gamma(a)$ and $\alpha(b, s) = \gamma(b)$ for every s ; equivalently, $V(a) = V(b) = 0$.

24 The first variation formulas

[Theorem] First variation of energy. For every variation of a smooth curve,

$$\frac{d}{ds} \Big|_{s=0} E(\gamma_s) = \langle V, \dot{\gamma} \rangle \Big|_a^b - \int_a^b \left\langle V, \frac{D\dot{\gamma}}{dt} \right\rangle dt.$$

For a fixed-endpoint variation, the boundary term vanishes.

[Proof] Since the coordinate vector fields of the variation commute and the connection is torsion-free,

$$\nabla_{\partial_s} T = \nabla_{\partial_t} \frac{\partial \alpha}{\partial s}.$$

Therefore

$$\begin{aligned} \frac{d}{ds} \Big|_{s=0} E(\gamma_s) &= \int_a^b \left\langle \frac{DV}{dt}, \dot{\gamma} \right\rangle dt \\ &= \langle V, \dot{\gamma} \rangle \Big|_a^b - \int_a^b \left\langle V, \frac{D\dot{\gamma}}{dt} \right\rangle dt. \end{aligned}$$

[Corollary] A smooth curve is critical for the energy under all fixed-endpoint variations if and only if it is an affinely parametrized geodesic.

[Theorem] First variation of length. Suppose γ is regular and put

$$U = \frac{\dot{\gamma}}{\|\dot{\gamma}\|}.$$

Then

$$\left. \frac{d}{ds} \right|_{s=0} L(\gamma_s) = \langle V, U \rangle|_a^b - \int_a^b \left\langle V, \frac{DU}{dt} \right\rangle dt.$$

In particular, the boundary term vanishes for fixed endpoints.

[Proposition] Suppose that γ is piecewise smooth with break points

$$a = t_0 < t_1 < \cdots < t_k < t_{k+1} = b.$$

For a piecewise smooth variation,

$$\begin{aligned} \left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = & - \int_a^b \left\langle V, \frac{D\dot{\gamma}}{dt} \right\rangle dt + \langle V, \dot{\gamma} \rangle|_a^b \\ & - \sum_{i=1}^k \left\langle V(t_i), \dot{\gamma}(t_i^+) - \dot{\gamma}(t_i^-) \right\rangle, \end{aligned}$$

where the integral is taken piecewise. For a regular curve, the analogous length formula is

$$\begin{aligned} \left. \frac{d}{ds} \right|_{s=0} L(\gamma_s) = & - \int_a^b \left\langle V, \frac{DU}{dt} \right\rangle dt + \langle V, U \rangle|_a^b \\ & - \sum_{i=1}^k \left\langle V(t_i), U(t_i^+) - U(t_i^-) \right\rangle. \end{aligned}$$

[Corollary] If a piecewise smooth curve is critical for energy under all fixed-endpoint variations, then it is C^1 and is a geodesic.

[Proof] Variations supported inside each smooth interval show that every piece is geodesic. The corner values $V(t_i)$ may be prescribed arbitrarily, so the corner terms force

$$\dot{\gamma}(t_i^+) = \dot{\gamma}(t_i^-).$$

Thus the pieces join with matching velocity and form one smooth geodesic.

25 The second variation formula and the index form

[Theorem] Second variation of energy. Let γ be a geodesic, let α be a variation of γ , and set

$$A = \nabla_{\partial_s} \frac{\partial \alpha}{\partial s} \Big|_{s=0}.$$

Then

$$\begin{aligned} \left. \frac{d^2}{ds^2} \right|_{s=0} E(\gamma_s) = & \left(\langle A, \dot{\gamma} \rangle + \left\langle V, \frac{DV}{dt} \right\rangle \right) \Big|_a^b \\ & + \int_a^b \left(\left\| \frac{DV}{dt} \right\|^2 - \langle \mathcal{R}(V, \dot{\gamma}) \dot{\gamma}, V \rangle \right) dt. \end{aligned}$$

Equivalently,

$$\left. \frac{d^2}{ds^2} \right|_{s=0} E(\gamma_s) = \left(\langle A, \dot{\gamma} \rangle + \left\langle V, \frac{DV}{dt} \right\rangle \right) \Big|_a^b - \int_a^b \left\langle V, \frac{D^2V}{dt^2} + \mathcal{R}(V, \dot{\gamma})\dot{\gamma} \right\rangle dt.$$

For a fixed-endpoint variation, all boundary terms vanish.

[Proof] Differentiate the first variation formula, commute the two covariant derivatives using $\mathcal{R}$, and use that $D\dot{\gamma}/dt = 0$. Integration by parts gives the second displayed form.

[Definition] For vector fields V, W along γ vanishing at the endpoints, the index form is

$$I(V, W) = \int_a^b \left(\left\langle \frac{DV}{dt}, \frac{DW}{dt} \right\rangle - \langle \mathcal{R}(V, \dot{\gamma})\dot{\gamma}, W \rangle \right) dt.$$

Thus the second variation of a fixed-endpoint variation is $I(V, V)$. The Jacobi equation

$$\frac{D^2V}{dt^2} + \mathcal{R}(V, \dot{\gamma})\dot{\gamma} = 0$$

is the Euler–Lagrange equation of the index form.

26 Synge’s theorem

[Lemma] The following elementary facts hold for orthogonal transformations:

1. Every element of $SO(2m + 1)$ has eigenvalue 1.
2. Every element $A \in O(2m)$ with $\det A = -1$ has eigenvalue 1.

[Proof] Nonreal eigenvalues occur in conjugate pairs, and each pair has product 1. The remaining real eigenvalues are 1 or -1 . The determinant and parity conditions force at least one eigenvalue to be 1.

[Remark] On a compact Riemannian manifold, every nontrivial free homotopy class contains a length-minimizing closed geodesic. If the manifold is nonorientable, one may similarly minimize among orientation-reversing closed curves.

[Theorem] Synge’s theorem. Let (M, g) be a compact connected Riemannian manifold with positive sectional curvature.

1. If $\dim M$ is even and M is orientable, then M is simply connected.
2. If $\dim M$ is odd, then M is orientable.

[Proof] For the first statement, suppose that $\dim M = 2m$ and that M is not simply connected. Choose a shortest noncontractible closed geodesic γ and parametrize it with unit speed. Let

$$P : T_{\gamma(0)}M \longrightarrow T_{\gamma(0)}M$$

be parallel transport around γ . Orientability gives $\det P = 1$, while

$$P\dot{\gamma}(0) = \dot{\gamma}(0).$$

Hence the restriction of P to the odd-dimensional space $\dot{\gamma}(0)^\perp$ lies in $SO(2m-1)$ and has a unit fixed vector X_0 . Parallel transport X_0 along γ . This produces a periodic parallel normal field X , and the corresponding closed variation satisfies

$$I(X, X) = - \int_a^b \langle \mathcal{R}(X, \dot{\gamma})\dot{\gamma}, X \rangle dt < 0.$$

This contradicts the minimizing property of γ .

For the second statement, suppose that $\dim M = 2m + 1$ and that M is nonorientable. Choose a shortest orientation-reversing closed geodesic γ . Its parallel transport has determinant -1 and fixes $\dot{\gamma}(0)$. Its restriction to the even-dimensional normal space therefore has determinant -1 , so the lemma again supplies a fixed normal vector. The resulting periodic parallel normal field gives the same strictly negative second variation, a contradiction.

[Corollary] If (M^{2m}, g) is compact, nonorientable, and has positive sectional curvature, then

$$\pi_1(M) \cong \mathbb{Z}_2.$$

[Proof] The oriented double cover $\widetilde{M} \rightarrow M$ is compact, orientable, and still has positive sectional curvature. Synge's theorem shows that $\widetilde{M}$ is simply connected. It is therefore the universal cover, whose deck group is $\mathbb{Z}_2$.

Day 6

27 The Hessian of the energy and the index form

Let $\gamma : [a, b] \rightarrow M$ be a geodesic. Consider a two-parameter fixed-endpoint variation

$$\alpha : [a, b] \times (-\varepsilon, \varepsilon)^2 \longrightarrow M, \quad \gamma_{s,r}(t) = \alpha(t, s, r),$$

with $\gamma_{0,0} = \gamma$, and write

$$X = \frac{\partial \alpha}{\partial s} \Big|_{(0,0)}, \quad Y = \frac{\partial \alpha}{\partial r} \Big|_{(0,0)}.$$

Polarizing the second variation formula gives

$$\frac{\partial^2}{\partial r \partial s} \Big|_{(0,0)} E(\gamma_{s,r}) = \int_a^b \left(\left\langle \frac{DX}{dt}, \frac{DY}{dt} \right\rangle - \langle \mathcal{R}(X, \dot{\gamma})\dot{\gamma}, Y \rangle \right) dt.$$

Let $\mathcal{V}_\gamma$ denote the vector space of continuous, piecewise smooth vector fields along γ , and let

$$\mathcal{V}_\gamma^0 = \{X \in \mathcal{V}_\gamma : X(a) = X(b) = 0\}.$$

[Definition] The **index form** of γ is the symmetric bilinear form

$$I(X, Y) = \int_a^b \left(\left\langle \frac{DX}{dt}, \frac{DY}{dt} \right\rangle - \langle \mathcal{R}(X, \dot{\gamma})\dot{\gamma}, Y \rangle \right) dt,$$

where the integral is evaluated on the smooth subintervals. It does not depend on the chosen subdivision.

Suppose

$$a = t_0 < t_1 < \cdots < t_k < t_{k+1} = b$$

contains all break points of X . Piecewise integration by parts gives

$$\begin{aligned} I(X, Y) = & - \int_a^b \left\langle \frac{D^2 X}{dt^2} + \mathcal{R}(X, \dot{\gamma})\dot{\gamma}, Y \right\rangle dt + \left\langle \frac{DX}{dt}, Y \right\rangle \Big|_a^b \\ & - \sum_{i=1}^k \left\langle \frac{DX}{dt}(t_i^+) - \frac{DX}{dt}(t_i^-), Y(t_i) \right\rangle. \end{aligned}$$

The integral is again taken piecewise. The last sum records the jumps in the covariant derivative of X .

[Lemma] A field $V \in \mathcal{V}_\gamma$ is a Jacobi field along γ if and only if

$$I(V, X) = 0 \quad \text{for every } X \in \mathcal{V}_\gamma^0.$$

[Proof] If V is Jacobi, then the Jacobi equation makes the integral term in the integration-by-parts formula vanish. A Jacobi field is smooth, so there are no jump terms, and the endpoint term vanishes because $X(a) = X(b) = 0$.

Conversely, assume that $I(V, X) = 0$ for every $X \in \mathcal{V}_\gamma^0$. On any smooth subinterval choose a nonnegative smooth function f with compact support and put

$$X = f \left(\frac{D^2 V}{dt^2} + \mathcal{R}(V, \dot{\gamma})\dot{\gamma} \right).$$

The integration-by-parts formula then yields

$$0 = - \int f \left\| \frac{D^2 V}{dt^2} + \mathcal{R}(V, \dot{\gamma})\dot{\gamma} \right\|^2 dt,$$

so V satisfies the Jacobi equation on every smooth subinterval. At a break point t_i , choose X supported near t_i and prescribe

$$X(t_i) = \frac{DV}{dt}(t_i^+) - \frac{DV}{dt}(t_i^-).$$

All integral terms vanish, and therefore

$$0 = - \left\| \frac{DV}{dt}(t_i^+) - \frac{DV}{dt}(t_i^-) \right\|^2.$$

Thus DV/dt is continuous at every break point. Hence V is C^1 , the Jacobi equation promotes it to a smooth field, and V is a Jacobi field on all of $[a, b]$.

[Remark] The usual multivariable second-derivative test has the following variational interpretation. If I is positive definite, then γ is a strict local minimizer of energy. If $I(X, X) < 0$ for some $X \in \mathcal{V}_\gamma^0$, a variation with variational field X decreases the energy. If I is merely nonnegative and has a nontrivial nullspace, second variation alone is inconclusive; the null directions are governed by Jacobi fields.

28 Conjugate points and minimizing geodesics

Let $p = \gamma(a)$ and $q = \gamma(b)$. Recall that $\gamma(t_0)$ is conjugate to p along γ if there is a nonzero Jacobi field J satisfying

$$J(a) = J(t_0) = 0.$$

[Theorem] Index theorem. Let $\gamma : [a, b] \rightarrow M$ be a geodesic, with $p = \gamma(a)$ and $q = \gamma(b)$, and let I be its index form on $\mathcal{V}_\gamma^0$.

1. The form I is positive definite if and only if p has no conjugate point on $\gamma((a, b])$. In this case, γ is locally length minimizing among piecewise smooth fixed-endpoint curves, uniquely up to orientation-preserving reparametrization.
2. There exists $X \in \mathcal{V}_\gamma^0$ with $I(X, X) < 0$ if and only if a conjugate point to p occurs at some $\gamma(t_0)$ with $a < t_0 < b$. In this case $\gamma|_{[a, b]}$ is not minimizing.
3. The form I is nonnegative but degenerate if and only if q is the first conjugate point to p along γ .

[Proof] We divide the proof according to the position of the first conjugate point.

Part 1: a conjugate point occurs in the open interval. Suppose that $\gamma(t_0)$ is conjugate to p for some $a < t_0 < b$. We construct an explicit negative direction for I .

Choose a nonzero Jacobi field J on $[a, t_0]$ satisfying

$$J(a) = J(t_0) = 0.$$

Then $DJ/dt(t_0) \neq 0$. Extend J by zero on $[t_0, b]$, and choose $Y \in \mathcal{V}_\gamma^0$ such that

$$Y(t_0) = -\frac{DJ}{dt}(t_0^-).$$

Writing $Y_1 = Y|_{[a, t_0]}$ and $Y_2 = Y|_{[t_0, b]}$, define, for $s > 0$,

$$Z_s(t) = \begin{cases} J(t) + sY_1(t), & a \leq t \leq t_0, \\ sY_2(t), & t_0 \leq t \leq b. \end{cases}$$

This is a continuous piecewise smooth field in $\mathcal{V}_\gamma^0$. Since J is Jacobi on $[a, t_0]$ and zero on $[t_0, b]$, integration by parts on the two sides of t_0 leaves only the corner term:

$$I(J, Y) = - \left\| \frac{DJ}{dt}(t_0^-) \right\|^2.$$

Consequently,

$$\begin{aligned} I(Z_s, Z_s) &= 2sI(J, Y) + s^2I(Y, Y) \\ &= -s \left(2 \left\| \frac{DJ}{dt}(t_0^-) \right\|^2 - sI(Y, Y) \right) < 0 \end{aligned}$$

for all sufficiently small $s > 0$. This proves the existence of a negative direction.

To deduce failure of minimality, fix such an s and choose a fixed-endpoint variation α_r with variational field Z_s . Then

$$\left. \frac{d}{dr} \right|_{r=0} E(\alpha_r) = 0, \quad \left. \frac{d^2}{dr^2} \right|_{r=0} E(\alpha_r) = I(Z_s, Z_s) < 0,$$

so $E(\alpha_r) < E(\gamma)$ for small nonzero r . Hence $\gamma|_{[a,b]}$ is not minimizing.

Part 2: the first conjugate point is the endpoint. Assume that q is the first conjugate point to p . We first prove that I is nonnegative.

Let $c < b$, and reparametrize $\gamma|_{[a,c]}$ on $[a, b]$ to obtain γ_c . Choose a parallel orthonormal frame $e_1, \dots, e_n$ along γ and write

$$X(t) = \sum_i x^i(t) e_i(t).$$

Define the corresponding field X^c along γ_c by rescaling the coefficient functions linearly from $[a, b]$ to $[a, c]$. Then $X^c \in \mathcal{V}_{\gamma_c}^0$ and $X^c \rightarrow X$ as $c \uparrow b$. Since γ_c contains no conjugate point to p , Part 4 below gives

$$I^{\gamma_c}(X^c, X^c) > 0$$

for $X \neq 0$. Passing to the limit as $c \uparrow b$ yields

$$I^\gamma(X, X) \geq 0 \quad \text{for every } X \in \mathcal{V}_\gamma^0.$$

We next prove degeneracy. Because p and q are conjugate, there is a nonzero Jacobi field J satisfying

$$J(a) = J(b) = 0.$$

Integration by parts and the Jacobi equation give

$$I(J, Y) = 0 \quad \text{for every } Y \in \mathcal{V}_\gamma^0.$$

Thus I is degenerate.

Part 3: converses.

We first record a consequence of nonnegativity that will also be used in Part 4. If $I(X, X) = 0$, then, for every $Y \in \mathcal{V}_\gamma^0$,

$$0 \leq I(X + tY, X + tY) = 2tI(X, Y) + t^2I(Y, Y)$$

for every $t \in \mathbb{R}$. Using both signs of sufficiently small t gives

$$I(X, Y) = 0 \quad \text{for every } Y \in \mathcal{V}_\gamma^0.$$

Now suppose that I is nonnegative and degenerate. There is a nonzero $X \in \mathcal{V}_\gamma^0$ such that $I(X, Y) = 0$ for every $Y \in \mathcal{V}_\gamma^0$. By the preceding Jacobi-field characterization, X is a Jacobi field, so q is conjugate to p . It is the first conjugate point: if an earlier conjugate point existed, Part 1 would give a field Z with $I(Z, Z) < 0$, contradicting nonnegativity.

Part 4: no conjugate point. Assume that there is no conjugate point on $\gamma((a, b])$. We prove that γ is locally minimizing and that I is positive definite.

Choose a partition

$$a = t_0 < t_1 < \cdots < t_k = b$$

so fine that each arc $\gamma([t_i, t_{i+1}])$ lies in a normal neighbourhood. Every sufficiently close fixed-endpoint curve $\tilde{\gamma}$ remains in the same normal neighbourhoods. Replacing its successive arcs by the unique short geodesics with the same endpoints and applying the local minimizing theorem from Day 5 gives

$$L(\tilde{\gamma}) \geq L(\gamma).$$

Equality is possible only for an orientation-preserving reparametrization of γ . Thus γ is locally length minimizing.

We now prove that I is positive definite. Since

$$E(\tilde{\gamma}) \geq \frac{L(\tilde{\gamma})^2}{2(b-a)} \geq \frac{L(\gamma)^2}{2(b-a)} = E(\gamma),$$

γ is also a local minimizer of energy. Therefore, for every $X \in \mathcal{V}_\gamma^0$, the second variation gives

$$I(X, X) \geq 0.$$

If I were not positive definite, it would have a nonzero null vector; by the observation in Part 3, I would then be degenerate. Part 3 would imply that q is conjugate to p , contrary to the hypothesis. Hence

$$I(X, X) > 0 \quad \text{for every } 0 \neq X \in \mathcal{V}_\gamma^0.$$

The three conjugate-point cases are mutually disjoint and exhaustive. Parts 1–4 therefore prove all three equivalences and complete the proof.

29 The fundamental index lemma

[Lemma] Fundamental index lemma. Suppose that p has no conjugate point along $\gamma((a, b])$. Let V be a Jacobi field along γ , and let $X \in \mathcal{V}_\gamma$ have the same endpoint values:

$$X(a) = V(a), \quad X(b) = V(b).$$

Then

$$I(V, V) \leq I(X, X),$$

with equality if and only if $X = V$.

[Proof] Put $W = X - V$. Then $W \in \mathcal{V}_\gamma^0$. Since V is Jacobi, the Jacobi-field characterization gives

$$I(V, W) = 0.$$

Therefore

$$I(X, X) = I(V + W, V + W) = I(V, V) + I(W, W).$$

Because there is no conjugate point to p along $\gamma((a, b])$, the index theorem says that I is positive definite on $\mathcal{V}_\gamma^0$. Hence $I(W, W) \geq 0$, with equality exactly when $W = 0$. This proves both the inequality and its equality statement.

30 Metric and geodesic completeness

[Definition] A Riemannian manifold (M, g) is **geodesically complete** if, for every $p \in M$ and every $v \in T_p M$, the geodesic determined by

$$\gamma(0) = p, \quad \dot{\gamma}(0) = v$$

is defined on all of $\mathbb{R}$.

[Definition] A metric space (M, d) is **complete** if every Cauchy sequence in M converges to a point of M .

We first prove the metric lemma used in Hopf–Rinow.

[Lemma] Let $p, q \in M$. Choose $\delta > 0$ small enough that the exponential map is a diffeomorphism on a neighbourhood of $\overline{B_\delta(0_p)}$ and

$$d(p, \exp_p(v)) = \|v\| \quad \text{for } \|v\| \leq \delta.$$

If $d(p, q) > \delta$, then there is a point $q' \in \partial B(p, \delta)$ such that

$$d(p, q) = \delta + d(q', q).$$

[Proof] Choose piecewise smooth curves σ_n from p to q such that

$$L(\sigma_n) \downarrow d(p, q).$$

Because $q \notin B(p, \delta)$, each σ_n has a first intersection p_n with $\partial B(p, \delta)$. The geodesic sphere is compact, since

$$\partial B(p, \delta) = \exp_p(\partial B_\delta(0_p))$$

is the continuous image of the compact Euclidean sphere $\partial B_\delta(0_p)$. After passing to a subsequence, assume that

$$p_n \longrightarrow q' \in \partial B(p, \delta).$$

Split σ_n at p_n . The part before p_n has length at least $d(p, p_n) = \delta$, while the part after p_n has length at least $d(p_n, q)$. Moreover,

$$d(q', q) \leq d(q', p_n) + d(p_n, q).$$

Therefore

$$L(\sigma_n) + d(p_n, q') \geq \delta + d(q', q).$$

Letting $n \rightarrow \infty$, using $L(\sigma_n) \rightarrow d(p, q)$ and $d(p_n, q') \rightarrow 0$, gives

$$d(p, q) \geq \delta + d(q', q).$$

The triangle inequality gives the reverse inequality,

$$d(p, q) \leq d(p, q') + d(q', q) = \delta + d(q', q),$$

and hence equality holds.

[Lemma] Fix $p \in M$ and suppose that every geodesic issuing from p is defined on $[0, \infty)$. Then every $q \in M$ can be joined to p by a minimizing geodesic.

[Proof] Put $r = d(p, q)$. The case $p = q$ is trivial, so assume $r > 0$. Choose $0 < \delta < r$ as in the preceding lemma. It gives a point

$$q_1 \in \partial B(p, \delta)$$

such that

$$r = \delta + d(q_1, q).$$

Let $\gamma : [0, \infty) \rightarrow M$ be the unit-speed radial geodesic from p through q_1 , so $\gamma(\delta) = q_1$. Define

$$A = \{t \in [0, r] : r = t + d(\gamma(t), q)\}.$$

The equality at q_1 , together with the triangle inequality along γ , shows that $[0, \delta] \subset A$. The set A is closed by continuity of the distance function. Let

$$t_0 = \sup A.$$

Suppose that $t_0 < r$. Choose $\varepsilon > 0$ smaller than both $r - t_0$ and the normal radius at $\gamma(t_0)$. Applying the preceding lemma with centre $\gamma(t_0)$ gives a point y satisfying

$$d(\gamma(t_0), y) = \varepsilon,$$

$$d(\gamma(t_0), q) = \varepsilon + d(y, q).$$

Because $t_0 \in A$,

$$r = t_0 + \varepsilon + d(y, q).$$

Let $\eta : [0, \varepsilon] \rightarrow M$ be the unique short minimizing geodesic from $\gamma(t_0)$ to y . The concatenation of $\gamma|_{[0, t_0]}$ and η has length $t_0 + \varepsilon$. Moreover,

$$\begin{aligned} r &\leq d(p, y) + d(y, q) \\ &\leq t_0 + \varepsilon + d(y, q) \\ &= r. \end{aligned}$$

Thus both inequalities are equalities, and the concatenated curve minimizes from p to y . A minimizing piecewise geodesic has no corner, by the first variation formula. Hence

$$\dot{\eta}(0) = \dot{\gamma}(t_0).$$

Uniqueness of geodesics with prescribed initial data now gives

$$\eta(s) = \gamma(t_0 + s), \quad 0 \leq s \leq \varepsilon.$$

Consequently $t_0 + \varepsilon \in A$, contradicting the definition of t_0 . Therefore $t_0 = r$, and

$$d(\gamma(r), q) = 0.$$

Thus $\gamma(r) = q$, and $\gamma|_{[0, r]}$ is a minimizing geodesic from p to q .

31 The Hopf–Rinow theorem

[Theorem] Hopf–Rinow theorem. Let (M, g) be a connected Riemannian manifold, let d be its Riemannian distance, and fix $p \in M$. The following are equivalent:

- (1) The metric space (M, d) is complete,
- (2) the Riemannian manifold (M, g) is geodesically complete,
- (3) every geodesic issuing from p is defined on all of $\mathbb{R}$.

When these conditions hold, every two points of M can be joined by a minimizing geodesic. Equivalently, for every $p \in M$, the map

$$\exp_p : T_p M \longrightarrow M$$

is defined everywhere and is surjective.

[Proof] The implication $2 \Rightarrow 3$ is immediate.

Assume 3. By the preceding lemma, every $x \in M$ can be joined to p by a minimizing geodesic. Let (x_n) be a Cauchy sequence and put

$$r_n = d(p, x_n).$$

Then

$$|r_n - r_m| \leq d(x_n, x_m),$$

so $r_n \rightarrow r$ for some $r \geq 0$. Write

$$x_n = \exp_p(r_n v_n), \quad v_n \in T_p M, \quad \|v_n\| = 1$$

when $x_n \neq p$. The unit sphere in $T_p M$ is compact, so some subsequence satisfies $v_{n_j} \rightarrow v$. Continuity of the exponential map gives

$$x_{n_j} = \exp_p(r_{n_j} v_{n_j}) \longrightarrow \exp_p(rv).$$

Since the original sequence is Cauchy, the whole sequence converges to the same point. Thus $3 \Rightarrow 1$.

Finally, assume 1 and let $\gamma : (a, b) \rightarrow M$ be a maximal geodesic. If $b < \infty$, choose $t_n \uparrow b$. Since γ has constant speed,

$$d(\gamma(t_n), \gamma(t_m)) \leq \|\dot{\gamma}\| |t_n - t_m|,$$

so $(\gamma(t_n))$ is Cauchy. Completeness gives a limit $x \in M$. In a coordinate neighbourhood of x , the velocity has constant Riemannian norm and hence bounded coordinate components. The geodesic equation and the standard continuation theorem for ordinary differential equations extend γ through time b , contradicting maximality. The same argument at the left endpoint shows that $a = -\infty$. Hence $1 \Rightarrow 2$.

The existence of minimizing geodesics now follows from the preceding lemma, applied at either endpoint. Surjectivity of $\exp_p$ follows because a minimizing geodesic from p to q has the form

$$t \longmapsto \exp_p(tv)$$

for a suitable $v \in T_p M$.

[Example] The flat cylinder $S^1 \times \mathbb{R}$ is complete and has no conjugate points. Every geodesic is locally minimizing, but a geodesic that winds sufficiently far around the circular factor need not be globally minimizing.

[Remark] Globally minimizing geodesics need not be unique. On the round sphere, antipodal points are joined by infinitely many minimizing geodesics.

[Example] The punctured plane

$$\mathbb{R}^2 \setminus \{0\}$$

with the Euclidean metric is incomplete. The sequence $((1/n, 0))$ is Cauchy but has no limit in the manifold, and the radial geodesic approaching the missing origin cannot be extended. The points $(-1, 0)$ and $(1, 0)$ have distance 2, but no curve in the punctured plane realizes this distance, so there is no minimizing geodesic between them.

32 Cut points and the cut locus

Let (M, g) be complete, fix $p \in M$, and let

$$\gamma : [0, \infty) \longrightarrow M$$

be a unit-speed geodesic with $\gamma(0) = p$.

[Definition] The **cut time** of γ is

$$t_0 = \sup\{t > 0 : \gamma|_{[0,t]} \text{ is minimizing}\}.$$

If $t_0 < \infty$, then $\gamma(t_0)$ is called a **cut point** of p along γ . The **cut locus** of p is

$$\text{Cut}(p) = \{q \in M : q \text{ is a cut point of } p\}.$$

Thus a radial geodesic minimizes up to its cut point and ceases to minimize beyond it.

[Examples]

(1) In Euclidean space $(\mathbb{R}^n, g_E)$,

$$\text{Cut}(p) = \emptyset.$$

Every two points are joined by a unique minimizing straight line.

(2) On the unit round sphere (S^n, g_{S^n}) ,

$$\text{Cut}(p) = \{-p\}.$$

Every geodesic issuing from p first ceases to minimize at time π , when it reaches the antipodal point $-p$.

(3) On the flat cylinder $S^1 \times \mathbb{R}$, if

$$p = (e^{i\theta_0}, z_0),$$

then

$$\text{Cut}(p) = \{(e^{i(\theta_0+\pi)}, z) : z \in \mathbb{R}\}.$$

Indeed, after lifting to $\mathbb{R}^2$, a point on this opposite vertical line has two equally short lifts, and hence two minimizing geodesics from p .

(4) In hyperbolic space $(\mathbb{H}^n, g_{\mathbb{H}})$,

$$\text{Cut}(p) = \emptyset.$$

Any two points are joined by a unique minimizing geodesic.

33 Uniqueness away from the cut locus

[Lemma] Let (M, g) be complete. If $q \notin \text{Cut}(p)$, then there is a unique minimizing geodesic from p to q .

[Proof] Existence follows from Hopf–Rinow. Suppose that

$$\gamma, \sigma : [0, \ell] \longrightarrow M$$

are unit-speed minimizing geodesics from p to q , where $\ell = d(p, q)$. Since q is not a cut point of p along γ , there is an $\varepsilon > 0$ such that $\gamma|_{[0, \ell+\varepsilon]}$ is still minimizing.

Concatenate σ with $\gamma|_{[\ell, \ell+\varepsilon]}$. Its length is $\ell + \varepsilon$, which equals

$$d(p, \gamma(\ell + \varepsilon)).$$

Hence the concatenated curve is minimizing. A minimizing piecewise geodesic has no corner, so

$$\dot{\sigma}(\ell) = \dot{\gamma}(\ell).$$

Uniqueness for the geodesic equation, solved backwards from q , gives $\sigma = \gamma$.

34 The first cut point

[Theorem] Let (M, g) be complete and let

$$\gamma : [0, \infty) \longrightarrow M$$

be a unit-speed geodesic with $p = \gamma(0)$. If $\gamma(t_0)$ is the first cut point of p along γ , then at least one of the following holds:

1. $\gamma(t_0)$ is the first conjugate point to p along γ ;
2. $\gamma(t_0)$ is the first point on γ that can be joined to p by two distinct minimizing geodesics.

[Proof] Choose $t_i \downarrow t_0$ with $t_i > t_0$. Since $\gamma|_{[0, t_i]}$ is not minimizing, Hopf–Rinow gives a unit-speed minimizing geodesic

$$\sigma_i : [0, s_i] \longrightarrow M$$

from p to $\gamma(t_i)$ with

$$s_i = d(p, \gamma(t_i)) < t_i.$$

Continuity of the distance function and the fact that $\gamma|_{[0, t_0]}$ is minimizing give

$$s_i \longrightarrow t_0.$$

Put $v_i = \dot{\sigma}_i(0)$. After passing to a subsequence, compactness of the unit sphere in $T_p M$ gives

$$v_i \longrightarrow v, \quad \|v\| = 1.$$

Therefore

$$\sigma(s) = \exp_p(sv), \quad 0 \leq s \leq t_0,$$

is a minimizing geodesic from p to $\gamma(t_0)$.

Let $u = \dot{\gamma}(0)$. There are two cases.

Case 1. Assume $v = u$. We have

$$\exp_p(s_i v_i) = \gamma(t_i) = \exp_p(t_i u),$$

where $s_i v_i \neq t_i u$, while both vectors converge to $t_0 u$. If $(d \exp_p)_{t_0 u}$ were invertible, the inverse function theorem would make $\exp_p$ injective on a neighbourhood of $t_0 u$, a contradiction. Thus $(d \exp_p)_{t_0 u}$ is singular, so $\gamma(t_0)$ is conjugate to p along γ . No earlier conjugate point can occur, because every segment $\gamma|_{[0, t]}$ with $t < t_0$ is minimizing, whereas the index theorem rules out minimization beyond an interior conjugate point. Hence $\gamma(t_0)$ is the first conjugate point.

Case 2. Assume $v \neq u$. Then σ and $\gamma|_{[0, t_0]}$ are distinct minimizing geodesics from p to $\gamma(t_0)$. For $t < t_0$, the point $\gamma(t)$ is not in $\text{Cut}(p)$, so the preceding lemma gives uniqueness there. Hence $\gamma(t_0)$ is the first point on γ joined to p by two distinct minimizing geodesics.

35 The injectivity domain of the exponential map

For a unit vector $u \in S_p M$, define its radial cut time by

$$c_p(u) = \sup\{t > 0 : s \mapsto \exp_p(su), 0 \leq s \leq t, \text{ is minimizing}\}.$$

[Definition] The **injectivity domain** at p is the star-shaped open set

$$\mathcal{I}_p = \{tu : u \in S_p M, 0 \leq t < c_p(u)\} \subset T_p M.$$

[Proposition] The restriction

$$\exp_p : \mathcal{I}_p \longrightarrow M \setminus \text{Cut}(p)$$

is a diffeomorphism.

[Proof] Every point outside $\text{Cut}(p)$ is joined to p by a unique minimizing geodesic, so the restricted exponential map is bijective. Before the cut time there is no conjugate point; hence $d\exp_p$ is invertible at every point of $\mathcal{I}_p$. The inverse function theorem gives a smooth local inverse everywhere, and these local inverses agree by injectivity. Therefore the restriction is a diffeomorphism.

36 Smoothness of the distance function

Fix $p \in M$ and write

$$r_p(x) = d(p, x).$$

[Theorem] The function r_p is smooth on

$$M \setminus (\text{Cut}(p) \cup \{p\}).$$

For every q in this set, let

$$\gamma : [0, r_p(q)] \longrightarrow M$$

be the unique unit-speed minimizing geodesic from p to q . Then

$$\text{grad } r_p|_q = \dot{\gamma}(r_p(q)).$$

[Proof] By the preceding proposition,

$$\exp_p : \mathcal{I}_p \setminus \{0_p\} \longrightarrow M \setminus (\text{Cut}(p) \cup \{p\})$$

is a diffeomorphism. On this set,

$$r_p(q) = \|\exp_p^{-1}(q)\|,$$

so r_p is smooth.

Now take $z \in T_q M$ and choose a smooth curve $q(s)$ with

$$q(0) = q, \quad q'(0) = z.$$

For small s , let γ_s be the unique minimizing geodesic from p to $q(s)$. These geodesics form a smooth variation of γ . The first variation formula gives

$$\begin{aligned} \langle z, \text{grad } r_p|_q \rangle &= \left. \frac{d}{ds} \right|_{s=0} r_p(q(s)) \\ &= \left. \frac{d}{ds} \right|_{s=0} L(\gamma_s) \\ &= \langle z, \dot{\gamma}(r_p(q)) \rangle. \end{aligned}$$

Since this holds for every $z \in T_q M$, the claimed gradient formula follows.

[Remark] The distance function is globally 1-Lipschitz:

$$|r_p(x) - r_p(y)| \leq d(x, y).$$

This follows immediately by applying the triangle inequality in both directions. Smoothness fails in general at p and on $\text{Cut}(p)$.

[Remark] The cut locus $\text{Cut}(p)$ has Riemannian measure zero. This is a deeper regularity fact about the radial cut-time function; it is not needed for the local arguments below.

37 The Hessian of the distance function

Let $q \notin \text{Cut}(p) \cup \{p\}$, put $\ell = r_p(q)$, and let $\gamma : [0, \ell] \rightarrow M$ be the unique unit-speed minimizing geodesic from p to q .

[Theorem] Let $x \in T_q M$ be perpendicular to $\dot{\gamma}(\ell)$, let $y \in T_q M$ be arbitrary, and let X be the unique Jacobi field along γ satisfying

$$X(0) = 0, \quad X(\ell) = x.$$

Then

$$(\text{Hess } r_p)_q(x, y) = \left\langle \frac{DX}{dt}(\ell), y \right\rangle.$$

[Proof] Choose a curve $q(s)$ in the geodesic sphere $r_p^{-1}(\ell)$ through q with $q'(0) = x$, and let γ_s be the unit-speed radial minimizing geodesic from p to $q(s)$ on $[0, \ell]$. Its variational field is X . Since the coordinate vector fields of the variation commute and the connection is torsion free,

$$\nabla_X \dot{\gamma} = \nabla_{\dot{\gamma}} X.$$

Using $\text{grad } r_p = \dot{\gamma}$ at the endpoint gives

$$\begin{aligned}
(\text{Hess } r_p)_q(x, y) &= \langle \nabla_x \text{grad } r_p, y \rangle \\
&= \langle \nabla_X \dot{\gamma}, y \rangle_{t=\ell} \\
&= \left\langle \frac{DX}{dt}(\ell), y \right\rangle.
\end{aligned}$$

[Corollary] The radial direction lies in the nullspace of the Hessian:

$$(\text{Hess } r_p)_q(\dot{\gamma}(\ell), y) = 0 \quad \text{for every } y \in T_q M.$$

For $x \perp \dot{\gamma}(\ell)$ and the Jacobi field X above,

$$(\text{Hess } r_p)_q(x, x) = \left\langle \frac{DX}{dt}(\ell), X(\ell) \right\rangle = I^\gamma(X, X).$$

[Proof] Since $|\text{grad } r_p| = 1$, differentiating its squared norm shows that the Hessian vanishes whenever one entry is radial. For the second identity, integrate the index form by parts. The Jacobi equation kills the integral term and $X(0) = 0$, leaving precisely the endpoint term at $t = \ell$.

38 The Cartan–Hadamard theorem

[Theorem] Cartan–Hadamard theorem. Let (M, g) be a complete connected Riemannian manifold with nonpositive sectional curvature.

1. For every $p \in M$, the exponential map $\exp_p : T_p M \rightarrow M$ is a covering map.
2. If M is simply connected, then $\exp_p$ is a diffeomorphism.

[Proof] No geodesic issuing from p has a conjugate point. Indeed, if $\gamma(t_0)$ were the first conjugate point, there would be a nonzero Jacobi field J with $J(0) = J(t_0) = 0$. The Jacobi equation gives $I(J, J) = 0$, whereas

$$I(J, J) = \int_0^{t_0} \left(\left\| \frac{DJ}{dt} \right\|^2 - \langle \mathcal{R}(J, \dot{\gamma})\dot{\gamma}, J \rangle \right) dt > 0$$

because the sectional curvature is nonpositive and J is nonzero. This contradiction shows that $d\exp_p$ is nonsingular everywhere, so $\exp_p$ is a local diffeomorphism.

It remains to pass from a local diffeomorphism to a covering map. When the sectional curvature is nonpositive, Jacobi-field comparison with Euclidean space gives

$$\|(d\exp_p)_v(w)\| \geq \|w\|$$

for every $v, w \in T_p M$ with $w \perp v$; equality holds in the radial direction by the Gauss lemma. Thus a lift through $\exp_p$ of a finite-length path in M has bounded Euclidean length in $T_p M$. Such a lift cannot escape to infinity in finite time, so the local lifting furnished by the inverse function theorem continues along the whole path. Therefore $\exp_p$ has the path-lifting property and is a covering map.

Finally, $T_p M$ is simply connected. If M is also simply connected, this covering has one sheet and is therefore a diffeomorphism.

39 Radial curvature and corresponding vectors

Let $\gamma : [0, a] \rightarrow M$ and $\bar{\gamma} : [0, a] \rightarrow \bar{M}$ be unit-speed geodesics on Riemannian manifolds of the same dimension. Define

$$K^-(t) = \min_{\substack{\Pi \subset T_{\gamma(t)} M \\ \dim \Pi = 2, \dot{\gamma}(t) \in \Pi}} \text{Sec}_M(\Pi)$$

and

$$\bar{K}^+(t) = \max_{\substack{\bar{\Pi} \subset T_{\bar{\gamma}(t)} \bar{M} \\ \dim \bar{\Pi} = 2, \dot{\bar{\gamma}}(t) \in \bar{\Pi}}} \text{Sec}_{\bar{M}}(\bar{\Pi}).$$

Thus K^- is the least radial sectional curvature along γ , while $\bar{K}^+$ is the greatest radial sectional curvature along $\bar{\gamma}$.

[Definition] Vectors $x \in T_{\gamma(t)} M$ and $\bar{x} \in T_{\bar{\gamma}(t)} \bar{M}$ are **corresponding**, written $x \sim \bar{x}$, if

$$\|x\| = \|\bar{x}\|, \quad \langle x, \dot{\gamma}(t) \rangle = \langle \bar{x}, \dot{\bar{\gamma}}(t) \rangle.$$

Parallel orthonormal frames with first vectors $\dot{\gamma}$ and $\dot{\bar{\gamma}}$ identify corresponding vectors by giving them the same coefficients.

[Lemma] Let X and $\bar{X}$ be Jacobi fields along γ and $\bar{\gamma}$ such that

$$X(0) = \bar{X}(0) = 0, \quad \left\langle \frac{DX}{dt}(0), \dot{\gamma}(0) \right\rangle = \left\langle \frac{D\bar{X}}{dt}(0), \dot{\bar{\gamma}}(0) \right\rangle.$$

If $X(a) \sim \bar{X}(a)$, then their perpendicular endpoint components also correspond.

[Proof] The tangential parts of Jacobi fields vanishing at 0 are

$$X^\top(t) = ct \dot{\gamma}(t), \quad \bar{X}^\top(t) = ct \dot{\bar{\gamma}}(t),$$

with the same constant c by the hypothesis on the initial derivatives. Thus the tangential components at a correspond. Subtracting them from the corresponding vectors $X(a)$ and $\bar{X}(a)$ proves the claim.

40 Index comparison

[Theorem] Index comparison theorem. Assume that γ has no conjugate point to $\gamma(0)$ on $\gamma((0, a])$ and

$$\bar{K}^+(t) \leq K^-(t) \quad 0 \leq t \leq a.$$

Let X and $\bar{X}$ be Jacobi fields along γ and $\bar{\gamma}$ such that

$$X(0) = \bar{X}(0) = 0, \quad X(a) \sim \bar{X}(a).$$

Then

$$I^\gamma(X, X) \leq I^{\bar{\gamma}}(\bar{X}, \bar{X}).$$

[Proof] The tangential Jacobi components are affine multiples of the velocity fields and agree under the correspondence, so it suffices to compare the perpendicular components. Choose parallel orthonormal frames

$$e_1 = \dot{\gamma}, e_2, \dots, e_n, \quad \bar{e}_1 = \dot{\bar{\gamma}}, \bar{e}_2, \dots, \bar{e}_n.$$

Write

$$\bar{X}(t) = \sum_{i=2}^n x^i(t) \bar{e}_i(t)$$

and define

$$Y(t) = \sum_{i=2}^n x^i(t) e_i(t).$$

Then $Y(0) = X(0)$ and $Y(a) = X(a)$. The fundamental index lemma gives

$$I^\gamma(X, X) \leq I^\gamma(Y, Y).$$

The derivative terms for Y and $\bar{X}$ agree. The radial curvature inequality gives

$$-\langle \mathcal{R}(Y, \dot{\gamma}) \dot{\gamma}, Y \rangle \leq -\langle \bar{\mathcal{R}}(\bar{X}, \dot{\bar{\gamma}}) \dot{\bar{\gamma}}, \bar{X} \rangle.$$

After integration,

$$I^\gamma(Y, Y) \leq I^{\bar{\gamma}}(\bar{X}, \bar{X}),$$

which proves the theorem.

41 Hessian comparison along radial geodesics

[Theorem] Local Hessian comparison. Let $\gamma : [0, a] \rightarrow M$ and $\bar{\gamma} : [0, a] \rightarrow \bar{M}$ be minimizing unit-speed geodesics starting at p and $\bar{p}$. Fix $b \in (0, a]$, put $q = \gamma(b)$ and $\bar{q} = \bar{\gamma}(b)$, and assume

$$q \notin \text{Cut}(p), \quad \bar{q} \notin \text{Cut}(\bar{p}).$$

Assume also that γ has no conjugate point on $(0, a]$ and

$$\bar{K}^+(t) \leq K^-(t) \quad 0 \leq t \leq a.$$

Choose corresponding vectors $x \in T_q M$ and $\bar{x} \in T_{\bar{q}} \bar{M}$. Then

$$(\text{Hess } r_p)_q(x, x) \leq (\text{Hess } \bar{r}_{\bar{p}})_{\bar{q}}(\bar{x}, \bar{x}).$$

[Proof] The radial parts of the two Hessians vanish, so it is enough to use the components of x and $\bar{x}$ perpendicular to the radial velocities. Let X and $\bar{X}$ be the Jacobi fields with zero initial values and these perpendicular endpoint values. The Hessian–index identity and index comparison give

$$\begin{aligned} (\text{Hess } r_p)_q(x, x) &= I^\gamma(X, X) \\ &\leq I^{\bar{\gamma}}(\bar{X}, \bar{X}) \\ &= (\text{Hess } \bar{r}_{\bar{p}})_{\bar{q}}(\bar{x}, \bar{x}). \end{aligned}$$

42 The Rauch comparison theorem

[Theorem] Rauch comparison theorem. Let X and $\bar{X}$ be Jacobi fields along unit-speed geodesics $\gamma : [0, a] \rightarrow M$ and $\bar{\gamma} : [0, a] \rightarrow \bar{M}$. Assume that

$$X(0) = \bar{X}(0) = 0, \quad \frac{DX}{dt}(0) \sim \frac{D\bar{X}}{dt}(0),$$

that $X \perp \dot{\gamma}$ and $\bar{X} \perp \dot{\bar{\gamma}}$, that γ has no conjugate point on $(0, a]$, and that

$$\bar{K}^+(t) \leq K^-(t) \quad 0 \leq t \leq a.$$

Then

$$\|X(t)\| \leq \|\bar{X}(t)\| \quad 0 \leq t \leq a.$$

Moreover, $\bar{\gamma}$ has no conjugate point to $\bar{\gamma}(0)$ on $(0, a]$.

[Proof] Put

$$\rho(t) = \|X(t)\|^2, \quad \bar{\rho}(t) = \|\bar{X}(t)\|^2.$$

Since γ has no conjugate point, $\rho(t) > 0$ for $t > 0$. The Jacobi equation gives

$$X(t) = t \frac{DX}{dt}(0) + O(t^3), \quad \bar{X}(t) = t \frac{D\bar{X}}{dt}(0) + O(t^3).$$

Hence

$$\lim_{t \downarrow 0} \frac{\bar{\rho}(t)}{\rho(t)} = 1.$$

Let c be the supremum of the times for which $\bar{X}$ is nonzero on $(0, c)$. For $0 < b < c$, normalize the endpoint values by

$$X_b(t) = \frac{X(t)}{\|X(b)\|}, \quad \bar{X}_b(t) = \frac{\bar{X}(t)}{\|\bar{X}(b)\|}.$$

Their endpoint values at b correspond. Index comparison and the endpoint formula for Jacobi fields give

$$\left\langle \frac{DX_b}{dt}(b), X_b(b) \right\rangle \leq \left\langle \frac{D\bar{X}_b}{dt}(b), \bar{X}_b(b) \right\rangle.$$

Equivalently,

$$\frac{1}{2} \frac{\rho'(b)}{\rho(b)} \leq \frac{1}{2} \frac{\bar{\rho}'(b)}{\bar{\rho}(b)}.$$

Therefore

$$\frac{d}{dt} \left(\frac{\bar{\rho}(t)}{\rho(t)} \right) \geq 0$$

on $(0, c)$. Together with the limit at zero, this yields $\bar{\rho}(t) \geq \rho(t)$. In particular, $\bar{X}$ cannot acquire a zero while X remains nonzero, so $c = a$. The desired inequality follows on all of $[0, a]$. Applying the same argument to every nonzero perpendicular initial derivative shows that $\bar{\gamma}$ has no conjugate point to $\bar{\gamma}(0)$ on $(0, a]$.

43 Comparison of differentials of exponential maps

[Theorem] Rauch comparison for exponential maps. Under the hypotheses of the Rauch comparison theorem, let

$$v = \dot{\gamma}(0), \quad \bar{v} = \dot{\bar{\gamma}}(0),$$

and choose corresponding vectors $w \in T_{\gamma(0)}M$ and $\bar{w} \in T_{\bar{\gamma}(0)}\bar{M}$, perpendicular to v and $\bar{v}$. Then, for $0 < t \leq a$,

$$\|(d \exp_{\gamma(0)})_{tv}(w)\| \leq \|(d \exp_{\bar{\gamma}(0)})_{t\bar{v}}(\bar{w})\|.$$

[Proof] The fields

$$J(t) = (d \exp_{\gamma(0)})_{tw}(tw), \quad \bar{J}(t) = (d \exp_{\bar{\gamma}(0)})_{t\bar{w}}(t\bar{w})$$

are Jacobi fields satisfying

$$J(0) = \bar{J}(0) = 0, \quad \frac{DJ}{dt}(0) = w, \quad \frac{D\bar{J}}{dt}(0) = \bar{w}.$$

The Rauch comparison inequality gives $\|J(t)\| \leq \|\bar{J}(t)\|$. Dividing by $t > 0$ and using linearity of the differentials in their vector arguments gives the displayed estimate.

Day 8

44 Constant-curvature model spaces

For $k \in \mathbb{R}$, let M_k^n denote the complete simply connected n -dimensional Riemannian manifold of constant sectional curvature k :

- the round sphere of radius $1/\sqrt{k}$ when $k > 0$;
- Euclidean space $\mathbb{R}^n$ when $k = 0$;
- hyperbolic space of curvature k when $k < 0$.

Define the model functions

$$\text{sn}_k(t) = \begin{cases} \frac{\sin(\sqrt{k}t)}{\sqrt{k}}, & k > 0, \\ t, & k = 0, \\ \frac{\sinh(\sqrt{-k}t)}{\sqrt{-k}}, & k < 0, \end{cases}$$

and

$$\text{cn}_k(t) := \text{sn}'_k(t) = \begin{cases} \cos(\sqrt{k}t), & k > 0, \\ 1, & k = 0, \\ \cosh(\sqrt{-k}t), & k < 0. \end{cases}$$

They satisfy

$$\text{sn}_k'' + k \text{sn}_k = 0, \quad \text{sn}_k(0) = 0, \quad \text{sn}'_k(0) = 1.$$

When $k > 0$, formulas involving $\text{cn}_k / \text{sn}_k$ are used only for $0 < t < \pi/\sqrt{k}$.

[Proposition] Let $r_p = d(p, \cdot)$ on M_k^n . Away from p and, when $k > 0$, its antipode,

$$\text{Hess } r_p = \frac{\text{cn}_k(r_p)}{\text{sn}_k(r_p)} (g - dr_p \otimes dr_p).$$

Equivalently, the radial eigenvalue is zero and all $n - 1$ tangential eigenvalues are

$$\frac{\text{cn}_k(r_p)}{\text{sn}_k(r_p)}.$$

[Proof] Let $\gamma : [0, \ell] \rightarrow M_k^n$ be the minimizing unit-speed geodesic from p to q , and let $x \in T_q M_k^n$ be perpendicular to $\dot{\gamma}(\ell)$. Parallel translate x along γ to a field $X(t)$. The Jacobi field with $J(0) = 0$ and $J(\ell) = x$ is

$$J(t) = \frac{\text{sn}_k(t)}{\text{sn}_k(\ell)} X(t).$$

The Hessian-index identity gives

$$\begin{aligned} (\text{Hess } r_p)_q(x, x) &= \langle J'(\ell), J(\ell) \rangle \\ &= \frac{\text{cn}_k(\ell)}{\text{sn}_k(\ell)} |x|^2. \end{aligned}$$

Since $\text{Hess } r_p$ vanishes in the radial direction, the tensor formula follows.

Define

$$S_k(t) := \int_0^t \operatorname{sn}_k(s) ds.$$

Thus

$$S'_k(t) = \operatorname{sn}_k(t), \quad S''_k(t) = \operatorname{cn}_k(t),$$

and explicitly

$$S_k(t) = \begin{cases} \frac{1 - \cos(\sqrt{k}t)}{k}, & k > 0, \\ \frac{t^2}{2}, & k = 0, \\ \frac{\cosh(\sqrt{-k}t) - 1}{-k}, & k < 0. \end{cases}$$

[Corollary] On M_k^n ,

$$\operatorname{Hess}(S_k(r_p)) = \operatorname{cn}_k(r_p) g.$$

[Proof] The chain rule and the preceding proposition give

$$\begin{aligned} \operatorname{Hess}(S_k(r_p)) &= \operatorname{sn}_k(r_p) \operatorname{Hess} r_p + \operatorname{cn}_k(r_p) dr_p \otimes dr_p \\ &= \operatorname{cn}_k(r_p) g. \end{aligned}$$

45 Barrier inequalities

Distance functions fail to be smooth on the cut locus. Barrier inequalities allow Hessian comparison to remain meaningful there.

[Definition] Let f be continuous on a Riemannian manifold M , and let $p \in M$. An **upper barrier** for f at p is a function $h \in C^2(U)$ on a neighbourhood U of p such that

$$h(p) = f(p), \quad h \geq f \quad \text{on } U.$$

A lower barrier is defined by reversing the inequality.

For $c \in \mathbb{R}$, the inequality

$$\Delta f(p) \leq c$$

holds **in the barrier sense** if, for every $\varepsilon > 0$, there is an upper barrier h_ε for f at p satisfying

$$\Delta h_\varepsilon(p) \leq c + \varepsilon.$$

More generally,

$$\operatorname{Hess} f \leq A$$

in the barrier sense if, for every $\varepsilon > 0$, there is an upper barrier h_ε at each point such that

$$\operatorname{Hess} h_\varepsilon \leq A + \varepsilon g$$

at that point.

[Lemma] Let $u : (a, b) \rightarrow \mathbb{R}$ be continuous.

- (1) If $u \in C^2$ and $u'' \leq C$ in the barrier sense, then its ordinary second derivative satisfies $u'' \leq C$.
- (2) If $u'' \leq C$ in the barrier sense and u attains an interior minimum, then $C \geq 0$.
- (3) If $-u'' \leq 0$ in the barrier sense, then u is convex.

[Proof] For the first assertion, if h is an upper barrier at t_0 , then $h - u$ has a local minimum at t_0 . Hence

$$u''(t_0) \leq h''(t_0) \leq C + \varepsilon,$$

and $\varepsilon \downarrow 0$ gives the result.

For the second assertion, let t_0 be an interior minimum of u . If $C < 0$, choose $\varepsilon > 0$ with $C + \varepsilon < 0$ and an upper barrier h at t_0 . For sufficiently small $\delta > 0$, Taylor's theorem gives

$$\frac{h(t_0 + \delta) + h(t_0 - \delta)}{2} < h(t_0) = u(t_0),$$

whereas $h \geq u \geq u(t_0)$ at both endpoints, a contradiction.

For the third assertion, suppose convexity fails. Then some affine function L agrees with u at two points while $u - L$ has a positive interior maximum. The function $L - u$ therefore has an interior minimum. Since $(L - u)'' \leq 0$ in the barrier sense, the second assertion forces equality; applying it after subtracting a small positive quadratic gives a contradiction. Thus u is convex.

[Remark] The Hessian inequality may be tested along geodesics. If $\sigma(0) = p$, then $\text{Hess } f \leq Cg$ in the barrier sense means that $f \circ \sigma$ has an upper C^2 support at 0 whose second derivative is at most $C|\dot{\sigma}(0)|^2 + \varepsilon$.

[Theorem] Hopf–Calabi strong maximum principle. Let $U \subset M$ be connected and open, and let $f \in C(U)$ satisfy

$$\Delta f \leq 0$$

in the barrier sense. If f attains its minimum at an interior point of U , then f is constant on U .

46 Global Hessian comparison

[Theorem] Global Hessian comparison. Let (M, g) be complete and assume

$$\text{Sec}_M \geq k.$$

Fix $p \in M$. On the set where $r_p < \pi/\sqrt{k}$ when $k > 0$, one has

$$\text{Hess}(S_k(r_p)) \leq \text{cn}_k(r_p) g$$

in the barrier sense.

[Proof] First suppose that $q \notin \{p\} \cup \text{Cut}(p)$. Let $\gamma : [0, \ell] \rightarrow M$ be the unique minimizing unit-speed geodesic from p to q . Local Hessian comparison with M_k^n gives

$$\text{Hess } r_p \leq \frac{\text{cn}_k(\ell)}{\text{sn}_k(\ell)} (g - dr_p \otimes dr_p)$$

at q . Therefore

$$\begin{aligned}\text{Hess}(S_k(r_p)) &= \text{sn}_k(r_p) \text{Hess } r_p + \text{cn}_k(r_p) dr_p \otimes dr_p \\ &\leq \text{cn}_k(r_p) g.\end{aligned}$$

At $q = p$, the function $S_k(r_p)$ is smooth near p . Indeed,

$$S_k(t) = \frac{t^2}{2} + O(t^4),$$

so in normal coordinates

$$S_k(r_p(x)) = \frac{1}{2}|x|^2 + O(|x|^4).$$

Consequently,

$$\text{Hess}(S_k(r_p))_p = g_p = \text{cn}_k(0)g_p.$$

It remains to treat $q \in \text{Cut}(p)$. Choose a minimizing unit-speed geodesic $\gamma : [0, \ell] \rightarrow M$ from p to q . For small $y > 0$, put

$$p_y = \gamma(y), \quad r_y(x) := y + d(p_y, x).$$

The triangle inequality gives

$$r_p(x) \leq r_y(x), \quad r_p(q) = r_y(q) = \ell.$$

For sufficiently small y , the function $d(p_y, \cdot)$ is smooth near q . Since S_k is increasing on the range under consideration,

$$h_y := S_k(r_y)$$

is an upper barrier for $S_k(r_p)$ at q .

Write $v = \text{grad } d(p_y, \cdot)|_q$. Local Hessian comparison, applied along the terminal segment of γ of length $\ell - y$, gives

$$\text{Hess } d(p_y, \cdot)_q \leq \frac{\text{cn}_k(\ell - y)}{\text{sn}_k(\ell - y)} (g - v^\flat \otimes v^\flat).$$

Hence

$$\begin{aligned}\text{Hess } h_y|_q &\leq \text{sn}_k(\ell) \frac{\text{cn}_k(\ell - y)}{\text{sn}_k(\ell - y)} (g - v^\flat \otimes v^\flat) \\ &\quad + \text{cn}_k(\ell) v^\flat \otimes v^\flat.\end{aligned}$$

The coefficient of the tangential part tends to $\text{cn}_k(\ell)$ as $y \downarrow 0$. Thus, for every $\varepsilon > 0$ and sufficiently small y ,

$$\text{Hess } h_y|_q \leq (\text{cn}_k(\ell) + \varepsilon)g_q.$$

This is precisely the desired barrier inequality.

47 Bonnet–Myers and maximal diameter

[Theorem] Bonnet–Myers theorem. Let (M^n, g) be complete. If there is a constant $k > 0$ such that

$$\text{Ric} \geq (n - 1)k g,$$

then M is compact and

$$\text{diam}(M) \leq \frac{\pi}{\sqrt{k}}.$$

[Proof] Let $\gamma : [0, \ell] \rightarrow M$ be any minimizing unit-speed geodesic. Choose parallel orthonormal vector fields $E_1, \dots, E_{n-1}$ perpendicular to $\dot{\gamma}$, and set

$$V_i(t) = \sin\left(\frac{\pi t}{\ell}\right) E_i(t).$$

Each V_i vanishes at the endpoints. Since γ is minimizing, the index form is nonnegative on such fields. Summing over i gives

$$\begin{aligned} 0 &\leq \sum_{i=1}^{n-1} I(V_i, V_i) \\ &= \int_0^\ell \left((n-1) \frac{\pi^2}{\ell^2} \cos^2\left(\frac{\pi t}{\ell}\right) - \text{Ric}(\dot{\gamma}, \dot{\gamma}) \sin^2\left(\frac{\pi t}{\ell}\right) \right) dt \\ &\leq \frac{n-1}{2} \left(\frac{\pi^2}{\ell} - k\ell \right). \end{aligned}$$

Therefore $\ell \leq \pi/\sqrt{k}$. Hopf–Rinow ensures that any two points are joined by a minimizing geodesic, so

$$\text{diam}(M) \leq \frac{\pi}{\sqrt{k}}.$$

A complete Riemannian manifold is a proper metric space; hence boundedness implies compactness.

[Theorem] Cheng maximal diameter theorem. Let (M^n, g) be complete and satisfy

$$\text{Ric} \geq (n-1)k g$$

for some $k > 0$. If

$$\text{diam}(M) = \frac{\pi}{\sqrt{k}},$$

then (M, g) is isometric to the round sphere of constant sectional curvature k .

48 Geodesic triangles and the model cosine law

[Definition] Let (M, g) be complete. A **geodesic triangle** $\triangle ABC$ consists of three vertices $A, B, C \in M$ together with chosen minimizing geodesic segments $\gamma_{AB}, \gamma_{BC}, \gamma_{CA}$ joining them. Its side lengths are

$$a = d(B, C), \quad b = d(C, A), \quad c = d(A, B),$$

so that a, b, c are opposite A, B, C , respectively.

[Definition] The **geodesic angle** at A is the angle between the initial velocities of the two chosen sides issuing from A :

$$\angle BAC = \angle(\dot{\gamma}_{AB}(0), \dot{\gamma}_{AC}(0)).$$

[Theorem] Model cosine law. Let $\triangle ABC$ be a geodesic triangle in the constant-curvature model M_k^n , with side lengths a, b, c opposite the corresponding angles A, B, C . Then

$$\operatorname{cn}_k(c) = \operatorname{cn}_k(a) \operatorname{cn}_k(b) + k \operatorname{sn}_k(a) \operatorname{sn}_k(b) \cos C.$$

[Proof] Let $\gamma_1 : [0, a] \rightarrow M_k^n$ be the side from C to B and $\gamma_2 : [0, b] \rightarrow M_k^n$ the side from C to A . Put

$$\phi(t) = S_k(r_B(\gamma_2(t))).$$

The model-space Hessian identity gives

$$\phi''(t) = 1 - k\phi(t).$$

Moreover,

$$\phi(0) = S_k(a), \quad \phi(b) = S_k(c),$$

and the first variation formula gives

$$\phi'(0) = -\operatorname{sn}_k(a) \cos C.$$

Since $\operatorname{cn}_k = 1 - kS_k$, the function

$$u(t) = 1 - k\phi(t)$$

satisfies $u'' + ku = 0$ with

$$u(0) = \operatorname{cn}_k(a), \quad u'(0) = k \operatorname{sn}_k(a) \cos C.$$

Solving this initial-value problem and evaluating at $t = b$ yields the formula.

49 Comparison triangles

[Definition] Let $\triangle ABC$ be a geodesic triangle in a complete Riemannian manifold. A **comparison triangle** in M_k^n is a geodesic triangle $\triangle \overline{ABC}$ satisfying

$$d(\overline{A}, \overline{B}) = d(A, B), \quad d(\overline{B}, \overline{C}) = d(B, C), \quad d(\overline{C}, \overline{A}) = d(C, A).$$

Its angles are called the comparison angles.

[Proposition] Suppose $a, b, c > 0$ satisfy the strict triangle inequalities. If $k > 0$, assume in addition that

$$a + b + c < \frac{2\pi}{\sqrt{k}}.$$

Then there is a geodesic triangle in M_k^n with side lengths a, b, c , unique up to an isometry of M_k^n .

[Proof] Fix two sides of lengths a and b issuing from one point and vary their included angle continuously from 0 to π . The distance between their endpoints varies continuously through every value permitted by the strict triangle inequalities. In the spherical case, the perimeter assumption keeps all three sides on the minimizing branches. The intermediate value theorem therefore gives the required third side. Uniqueness follows from the model cosine law, which determines the included angle, together with the homogeneity and isotropy of M_k^n .

50 Toponogov comparison

[Theorem] Toponogov comparison theorem. Let (M, g) be complete and assume

$$\text{Sec}_M \geq k.$$

(1) Let $\gamma_0 : [0, b] \rightarrow M$ and $\gamma_1 : [0, c] \rightarrow M$ be minimizing unit-speed geodesics with the same initial point A . Let the corresponding model hinge in M_k^n have the same side lengths and the same included angle. If $B = \gamma_0(b)$ and $C = \gamma_1(c)$, then

$$d(B, C) \leq d(\overline{B}, \overline{C}).$$

(2) Let $\triangle ABC$ be a geodesic triangle and let $\triangle \overline{ABC} \subset M_k^n$ be its comparison triangle. Then

$$\angle A \geq \angle \overline{A}, \quad \angle B \geq \angle \overline{B}, \quad \angle C \geq \angle \overline{C}.$$

When $k > 0$, the statements are understood under the standard restriction that the relevant comparison triangle has perimeter less than $2\pi/\sqrt{k}$.

[Lemma] Let $f : [0, \ell] \rightarrow \mathbb{R}$ be continuous, with a right derivative at 0, and suppose

$$f(0) = 0, \quad f'_+(0) \leq 0, \quad f'' + kf \leq 0$$

in the barrier sense on $(0, \ell)$. If $k > 0$, also assume $\ell < \pi/\sqrt{k}$. Then

$$f(t) \leq 0 \quad \text{for every } t \in [0, \ell].$$

[Proof] For $\varepsilon > 0$, set

$$f_\varepsilon(t) = f(t) - \varepsilon \text{sn}_k(t).$$

Then

$$f_\varepsilon(0) = 0, \quad (f_\varepsilon)'_+(0) < 0, \quad f''_\varepsilon + kf_\varepsilon \leq 0$$

in the barrier sense. Thus $f_\varepsilon < 0$ for all sufficiently small positive t . Suppose that it has a first zero $t_0 > 0$.

First let $k < 0$. The function f_ε attains a negative minimum at some $t_1 \in (0, t_0)$. At that point the differential inequality gives

$$f''_\varepsilon(t_1) \leq -kf_\varepsilon(t_1) < 0$$

in the barrier sense, contradicting the barrier second-derivative test at a strict interior minimum.

If $k = 0$, then $f''_\varepsilon \leq 0$ in the barrier sense. The one-dimensional barrier lemma from Section 45, applied to $-f_\varepsilon$, shows that f_ε is concave. Since it vanishes at 0 and t_0 , it must be nonnegative between them, contradicting $f_\varepsilon < 0$ on $(0, t_0)$.

Now let $k > 0$. Choose $\delta > 0$ and $\eta > 0$ so small that

$$t_0 + \eta < \frac{\pi}{\sqrt{k + \delta}},$$

and define

$$\varphi(t) = -\sin(\sqrt{k + \delta}(t + \eta)).$$

Then $\varphi < 0$ on $[0, t_0]$ and

$$\varphi'' + (k + \delta)\varphi = 0.$$

The quotient f_ε/φ is zero at both endpoints and positive inside, so it has a positive interior maximum at some t_1 . Let h be an upper barrier for f_ε at t_1 . Because $\varphi < 0$, the quotient h/φ has a local maximum at t_1 . Choosing the barrier error smaller than $-\delta f_\varepsilon(t_1)$ gives

$$h''(t_1) + (k + \delta)h(t_1) < 0.$$

At the critical point of the quotient,

$$\left(\frac{h}{\varphi}\right)''(t_1) = \frac{h''(t_1) + (k + \delta)h(t_1)}{\varphi(t_1)} > 0,$$

contradicting the local maximum. Hence no first zero exists and $f_\varepsilon < 0$ on $(0, \ell]$. Letting $\varepsilon \downarrow 0$ proves the claim.

[Proof] We first prove the hinge statement. Write

$$\alpha = \angle(\dot{\gamma}_0(0), \dot{\gamma}_1(0)).$$

For small $\varepsilon > 0$, define

$$\rho_\varepsilon(t) = d(\gamma_0(b - \varepsilon), \gamma_1(t)), \quad 0 \leq t \leq c,$$

and define $\bar{\rho}_\varepsilon$ in the model hinge in the same way. Global Hessian comparison, applied along γ_1 , gives

$$(S_k \circ \rho_\varepsilon)'' \leq 1 - kS_k \circ \rho_\varepsilon$$

in the barrier sense. In the model space equality holds for $S_k \circ \bar{\rho}_\varepsilon$. Consequently,

$$F_\varepsilon = S_k \circ \rho_\varepsilon - S_k \circ \bar{\rho}_\varepsilon$$

satisfies

$$F_\varepsilon'' + kF_\varepsilon \leq 0$$

in the barrier sense. The two hinges have the same radial side and angle, so the first variation formula gives

$$F_\varepsilon(0) = 0, \quad F_\varepsilon'(0) = 0.$$

The lemma yields $F_\varepsilon \leq 0$. Evaluating at $t = c$, using the monotonicity of S_k on the relevant range, and then letting $\varepsilon \downarrow 0$ gives

$$d(B, C) \leq d(\bar{B}, \bar{C}).$$

For the triangle statement, construct a model hinge with the same two sides adjacent to A and with included angle $\angle A$. Part (1) says that its opposite side is at least $d(B, C)$, which is the opposite side of the comparison triangle. The model cosine law is monotone in the included angle, so

$$\angle A \geq \angle \bar{A}.$$

The same argument at the other two vertices proves the remaining angle inequalities.

Day 9

51 Ricci comparison along a minimizing geodesic

Let

$$\text{sn}_k(t) = \begin{cases} \frac{1}{\sqrt{k}} \sin(\sqrt{k} t), & k > 0, \\ t, & k = 0, \\ \frac{1}{\sqrt{-k}} \sinh(\sqrt{-k} t), & k < 0. \end{cases}$$

[Theorem] Ricci comparison. Let (M^n, g) be a Riemannian manifold, let $\gamma : [0, b] \rightarrow M$ be a minimizing unit-speed geodesic with $\gamma(0) = p$, and suppose

$$\text{Ric}(\dot{\gamma}(t), \dot{\gamma}(t)) \geq (n-1)k$$

for every $t \in [0, b]$. Then, at every point $\gamma(t)$ before the cut locus of p ,

$$\Delta r_p(\gamma(t)) \leq (n-1) \frac{\text{sn}'_k(t)}{\text{sn}_k(t)}.$$

Equivalently, the Laplacian of r_p is bounded above by the corresponding radial Laplacian in the constant-curvature model M_k^n .

More explicitly, if $\bar{\gamma}$ is any unit-speed geodesic in M_k^n with $\bar{\gamma}(0) = \bar{p}$, then

$$\Delta r_p(\gamma(t)) \leq \bar{\Delta} r_{\bar{p}}(\bar{\gamma}(t)).$$

Equality along $[0, b]$ holds precisely when

$$\text{Ric}(\dot{\gamma}(t), \dot{\gamma}(t)) = (n-1)k$$

and the transverse endpoint Jacobi fields are the model fields

$$X_i(s) = \frac{\text{sn}_k(s)}{\text{sn}_k(t)} e_i(s), \quad i = 2, \dots, n.$$

[Proof] Fix $t \in (0, b]$ and choose a parallel orthonormal frame

$$e_1 = \dot{\gamma}, e_2, \dots, e_n$$

along γ . For each $i \geq 2$, let X_i be the Jacobi field satisfying

$$X_i(0) = 0, \quad X_i(t) = e_i(t).$$

The Hessian-index lemma gives

$$(\text{Hess } r_p)_{\gamma(t)}(e_i(t), e_i(t)) = I(X_i, X_i).$$

Since $\text{Hess } r_p(\dot{\gamma}, \dot{\gamma}) = 0$, it follows that

$$\Delta r_p(\gamma(t)) = \sum_{i=2}^n I(X_i, X_i).$$

Use the comparison fields

$$\tilde{X}_i(s) = \frac{\text{sn}_k(s)}{\text{sn}_k(t)} e_i(s), \quad 0 \leq s \leq t.$$

They have the same endpoint values as X_i . The index lemma therefore implies

$$I(X_i, X_i) \leq I(\tilde{X}_i, \tilde{X}_i).$$

Summing over $i = 2, \dots, n$ gives

$$\begin{aligned} \Delta r_p(\gamma(t)) &\leq \int_0^t \left\{ (n-1) \left(\frac{\text{sn}'_k(s)}{\text{sn}_k(t)} \right)^2 - \left(\frac{\text{sn}_k(s)}{\text{sn}_k(t)} \right)^2 \text{Ric}(\dot{\gamma}(s), \dot{\gamma}(s)) \right\} ds \\ &\leq (n-1) \int_0^t \left\{ \left(\frac{\text{sn}'_k(s)}{\text{sn}_k(t)} \right)^2 - k \left(\frac{\text{sn}_k(s)}{\text{sn}_k(t)} \right)^2 \right\} ds. \end{aligned}$$

Because $\text{sn}_k'' + k \text{sn}_k = 0$, the integrand is the derivative of

$$\frac{\text{sn}_k(s) \text{sn}_k'(s)}{\text{sn}_k(t)^2}.$$

Consequently,

$$\Delta r_p(\gamma(t)) \leq (n-1) \frac{\text{sn}_k'(t)}{\text{sn}_k(t)}.$$

Equality at a fixed t forces equality in both comparison steps. Thus $X_i = \tilde{X}_i$ for every $i \geq 2$ and

$$\text{Ric}(\dot{\gamma}, \dot{\gamma}) = (n-1)k$$

along the segment. The Jacobi equation then gives

$$R(e_i, \dot{\gamma})\dot{\gamma} = k e_i, \quad i = 2, \dots, n,$$

so every radial sectional curvature along the segment equals k .

52 Riemannian volume in polar coordinates

If $\Sigma \subset \mathbb{R}^3$ is parametrized by $F(u, v) = (x(u, v), y(u, v), z(u, v))$, its area is

$$\text{Area}(\Sigma) = \int |F_u \wedge F_v| du dv.$$

For an n -dimensional Riemannian manifold this becomes

$$\text{Vol}(U) = \int_{\varphi(U)} \sqrt{\det(g_{ij} \circ \varphi^{-1})} dx^1 \cdots dx^n$$

in a coordinate chart φ .

Indeed, if $\partial_i = a_i^j e_j$ relative to an orthonormal frame, then

$$g_{ij} = \langle \partial_i, \partial_j \rangle = (AA^T)_{ij},$$

and hence

$$\sqrt{\det(g_{ij})} = |\det A|.$$

This is exactly the factor by which coordinate volume differs from Riemannian volume.

For a set not contained in one coordinate chart, choose a partition of unity $\{\rho_\alpha\}$ subordinate to coordinate charts $\{(U_\alpha, \varphi_\alpha)\}$. Then

$$\text{Vol}(U) = \sum_\alpha \int_{\varphi_\alpha(U \cap U_\alpha)} (\rho_\alpha \sqrt{\det g}) \circ \varphi_\alpha^{-1} dx^1 \cdots dx^n.$$

Fix $p \in M$. Since M is complete, the complement of the injectivity domain of $\exp_p$ and the cut locus of p have measure zero. We may therefore use geodesic polar coordinates

$$(r, \theta) \mapsto \exp_p(r\theta), \quad \theta \in S_p M.$$

After identifying $T_p M$ with $\mathbb{R}^n$, Euclidean polar measure is

$$dx^1 \cdots dx^n = r^{n-1} dr d\theta.$$

For $\theta \in S_p M$, let

$$\gamma_\theta(r) = \exp_p(r\theta)$$

and choose an orthonormal basis $v_2, \dots, v_n$ of $\theta^\perp \subset T_p M$. Let V_i be the Jacobi field along γ_θ determined by

$$V_i(0) = 0, \quad V_i'(0) = v_i.$$

Equivalently,

$$V_i(r) = d(\exp_p)_{r\theta}(rv_i).$$

Let $y^2, \dots, y^n$ be local coordinates on $S_p M$ chosen so that $\partial/\partial y^i = v_i$ at θ . The corresponding polar coordinate vectors are

$$\frac{\partial}{\partial r} = \dot{\gamma}_\theta(r), \quad \frac{\partial}{\partial y^i} = V_i(r).$$

Gauss's lemma gives

$$\left\langle \frac{\partial}{\partial r}, V_i(r) \right\rangle = 0, \quad \left\| \frac{\partial}{\partial r} \right\| = 1.$$

Thus the determinant of the polar-coordinate metric is

$$\det(g_{ab}) = \det(\langle V_i(r), V_j(r) \rangle)_{i,j=2}^n.$$

[Definition] Before the cut time of θ , define the radial volume density by

$$\mu_p(r, \theta) = \sqrt{\det(\langle V_i(r), V_j(r) \rangle)_{i,j=2}^n}.$$

Equivalently, if the angular basis is not orthonormal, then

$$\mu_p(r, \theta) = \frac{|\det(V_2(r), \dots, V_n(r))|}{|\det(v_2, \dots, v_n)|}.$$

This quotient shows that the definition is independent of the chosen angular coordinates. At and beyond the cut time, set $\mu_p(r, \theta) = 0$. Then

$$d \operatorname{vol}_g = \mu_p(r, \theta) dr d\theta,$$

and hence

$$\begin{aligned} \operatorname{Area}(S(p, r)) &= \int_{S_p M} \mu_p(r, \theta) d\theta, \\ \operatorname{Vol}(B(p, r)) &= \int_0^r \int_{S_p M} \mu_p(t, \theta) d\theta dt. \end{aligned}$$

Before the cut time, write

$$A(r) = (\langle V_i(r), V_j(r) \rangle)_{i,j=2}^n.$$

Since $\mu_p(r, \theta) = \sqrt{\det A(r)}$, Jacobi's determinant formula gives

$$\frac{d}{dr} \log \mu_p(r, \theta) = \frac{1}{2} \operatorname{tr}(A(r)^{-1} A'(r)) = \Delta r_p(\gamma_\theta(r)).$$

Here is the last equality in detail. At a fixed regular value r , change the angular basis so that

$V_2(r), \dots, V_n(r)$ is orthonormal. Then $A(r) = \text{Id}$ and

$$\begin{aligned} \frac{1}{2} \text{tr } A'(r) &= \sum_{i=2}^n \langle V_i'(r), V_i(r) \rangle \\ &= \sum_{i=2}^n I(V_i, V_i) \\ &= \sum_{i=2}^n (\text{Hess } r_p)_{\gamma_\theta(r)}(V_i(r), V_i(r)) \\ &= \Delta r_p(\gamma_\theta(r)). \end{aligned}$$

In the model space M_k^n , the corresponding density is

$$\mu_k(r) = \text{sn}_k(r)^{n-1}.$$

53 Bishop–Gromov volume comparison

[Proposition] Let (M^n, g) be complete and suppose

$$\text{Ric}_M \geq (n-1)k g.$$

For every $p \in M$ and $\theta \in S_p M$, the function

$$r \mapsto \frac{\mu_p(r, \theta)}{\text{sn}_k(r)^{n-1}}$$

is nonincreasing wherever the denominator is positive. Moreover,

$$\lim_{r \downarrow 0} \frac{\mu_p(r, \theta)}{\text{sn}_k(r)^{n-1}} = 1.$$

Consequently,

$$\mu_p(r, \theta) \leq \text{sn}_k(r)^{n-1}.$$

[Proof] Before the cut time, Ricci comparison gives

$$\begin{aligned} \frac{d}{dr} \log \frac{\mu_p(r, \theta)}{\text{sn}_k(r)^{n-1}} &= \Delta r_p(\gamma_\theta(r)) - (n-1) \frac{\text{sn}_k'(r)}{\text{sn}_k(r)} \\ &\leq 0. \end{aligned}$$

In normal coordinates, the Jacobi fields satisfy

$$V_i(r) = r v_i + O(r^3),$$

so both the numerator and denominator are asymptotic to r^{n-1} . This proves the limit. Extending the numerator by zero beyond the cut time preserves monotonicity.

The equality case can also be read directly from this proof. If

$$\mu_p(t, \theta) = \text{sn}_k(t)^{n-1}$$

throughout an interval, then the logarithmic derivative inequality is an equality there. Hence

$$\Delta r_p(\gamma_\theta(t)) = (n-1) \frac{\text{sn}_k'(t)}{\text{sn}_k(t)}.$$

Equality in Ricci comparison gives

$$V_i(t) = \operatorname{sn}_k(t)e_i(t)$$

for a parallel orthonormal frame $e_2, \dots, e_n$, and therefore

$$R(e_i, \dot{\gamma}_\theta)\dot{\gamma}_\theta = ke_i.$$

[Theorem] Bishop–Gromov comparison. Under the same Ricci lower bound, let $B_k(r)$ and $S_k(r)$ denote the radius- r ball and sphere in the simply connected space form of curvature k . Then

$$r \mapsto \frac{\operatorname{Area}(S(p, r))}{\operatorname{Area}(S_k(r))}$$

and

$$r \mapsto \frac{\operatorname{Vol}(B(p, r))}{\operatorname{Vol}(B_k(r))}$$

are nonincreasing wherever the model quantities are defined. Both ratios tend to 1 as $r \downarrow 0$. In particular,

$$\operatorname{Vol}(B(p, r)) \leq \operatorname{Vol}(B_k(r)).$$

[Lemma] Let $f, s : [0, \infty) \rightarrow \mathbb{R}$ with $s > 0$. If f/s is nonincreasing, then

$$r \mapsto \frac{\int_0^r f(t) dt}{\int_0^r s(t) dt}$$

is also nonincreasing.

[Proof] Differentiating the ratio gives

$$\frac{s(r)}{(\int_0^r s)^2} \int_0^r \left(\frac{f(r)}{s(r)} - \frac{f(t)}{s(t)} \right) s(t) dt \leq 0.$$

[Proof] Integrating the pointwise density comparison over $S_p M \cong S^{n-1}$ proves that the sphere-area ratio is nonincreasing. Apply this with

$$f(r) = \operatorname{Area}(S(p, r)), \quad s(r) = \operatorname{Area}(S_k(r)).$$

The lemma gives the asserted monotonicity of the ball-volume ratio. Normal coordinates show that both the area and volume ratios tend to 1 as $r \downarrow 0$.

[Remark] Rigidity. If equality in the Bishop–Gromov ball comparison holds at some radius R and there is no loss through the cut locus before R , then every radial density agrees with the model density on $(0, R)$. Equality in the Ricci comparison proof forces every radial sectional curvature to be k . The exponential map therefore identifies $B(p, R)$ isometrically with the radius- R model ball. More generally, equality of the volume ratio at two radii forces the annulus between them to be the corresponding model annulus.

54 Maximal diameter rigidity

[Theorem] Cheng's maximal diameter theorem. Let (M^n, g) be complete and suppose

$$\text{Ric}_M \geq (n-1)k g, \quad k > 0.$$

If

$$\text{diam}(M) = \frac{\pi}{\sqrt{k}},$$

then (M, g) is isometric to the round sphere of constant curvature k .

[Proof] Bonnet–Myers implies that M is compact. Choose $p, q \in M$ with

$$d(p, q) = \frac{\pi}{\sqrt{k}}.$$

Set $R = \pi/(2\sqrt{k})$. The open balls $B(p, R)$ and $B(q, R)$ are disjoint, while $B(p, 2R) = M$. Bishop–Gromov monotonicity gives

$$\frac{\text{Vol}(B(p, R))}{\text{Vol}(B_k(R))} \geq \frac{\text{Vol}(M)}{\text{Vol}(S_k^n)}.$$

Since a hemisphere has half the volume of the round sphere,

$$\text{Vol}(B(p, R)) \geq \frac{1}{2} \text{Vol}(M).$$

The same argument at q gives the same lower bound for $B(q, R)$. Their disjointness forces equality throughout. The rigidity case of Bishop–Gromov therefore identifies both half-balls with model hemispheres, and their boundaries glue by the radial identification. Thus (M, g) is isometric to the round sphere S_k^n .

[Remark] A further application of comparison geometry is to the fundamental group. Choose shortest geodesic loops representing a minimal set of generators of $\pi_1(M, p)$. Comparison estimates give a definite lower bound for the angles between their initial directions. The resulting spherical-packing estimate on $S_p M$ bounds the number of such generators.